CSI2TEXT-PGMT: Physics-Grounded Multitask Channel Reporting from CSI

***Abstract*—Raw channel state information (CSI) is high-dimensional and complex-valued, making it difficult for network controllers and operators to directly interpret, retrieve and reuse. Future autonomous wireless networks require not only channel state estimation but also the understanding of underlying physical propagation mechanisms. Although existing unconstrained language decoders can generate fluent reports, their outputs may violate physical facts. To address this issue, this paper proposes CSI2TEXT-PGMT, a physics-prior-constrained multi-task framework that transforms heterogeneous beam-frequency CSI into structured and verifiable channel descriptions. The system-configuration-aware Transformer encoder learns shared CSI representations via semantic alignment with textual descriptions and propagation prototypes. Multiple task-specific branches estimate propagation attributes concerning delay, angle, power and multipath structure. A relation correction module enforces physical dependencies among diverse predictions, while a deterministic language generation module produces final channel analysis reports. Extensive experiments verify that the framework accurately recovers various propagation attributes, distinguishes LoS/NLoS scenarios, reduces logical inconsistencies among joint physical predictions, and generates physically faithful descriptions. Furthermore, learned representations are evaluated on downstream tasks including adaptive beam search, MCS link adaptation and link quality/outage prediction, which demonstrates the framework's capability for physical channel interpretation and effective support for wireless resource management decisions.**

***Index Terms*—Channel state information, channel reporting, physics-grounded learning, wireless intelligence, multitask learning.**

I. Introduction

Autonomous wireless networks require more than accurate channel-state estimation; they also need interpretable knowledge of the underlying propagation conditions, including whether a link is LoS or NLoS, which path dominates, how severe the delay and angular dispersion are, and whether reflection-rich propagation may impair beam management or positioning. However, raw CSI does not directly provide such an interface: conventional estimators return fragmented physical attributes, whereas unconstrained language decoders may produce plausible descriptions that are not supported by the measured channel.

Recent studies have made important progress in learning representations from wireless channels and connecting wireless signals with language. Channel charting recovers geometry-preserving representations from CSI [18], while reusable CSI representations support channel prediction, positioning, and cross-configuration transfer [9], [11]. More recently, wireless-language models have generated human-action captions from Wi-Fi CSI [20], aligned unsegmented CSI streams with language models for zero-shot activity understanding [8], and introduced an RF-language model for waveform and spectrum understanding [23]. These methods demonstrate the potential of wireless-language modeling, but they primarily describe human activities or communication-signal content rather than the propagation channel itself. Meanwhile, existing CSI models usually output latent embeddings, reconstructed channels, locations, or selected physical parameters, leaving propagation information fragmented and inaccessible as a unified channel report.

This separation reveals a fundamental challenge. CSI-to-language conversion requires a model to first recover a coherent physical record and then express that record in natural language. The first stage determines whether quantities such as delay, angle, power, and path count are supported by the CSI, whereas the second determines whether they are presented completely and readably. Assigning both responsibilities to an unconstrained language decoder can produce fluent and structurally plausible descriptions without guaranteeing numerical accuracy or physical consistency. For propagation-aware network operation, such errors are critical because a well-written but incorrect channel report may directly mislead downstream decisions.

In this work, we take a different perspective: rather than asking a large language model to freely interpret CSI, we formulate CSI-to-text conversion as physics-grounded structured inference followed by faithful verbalization. We propose CSI2Text-PGMT, a configuration-aware multitask framework that transforms beam-frequency CSI into an auditable channel description. A shared Transformer encoder processes phase-aligned and beamspace-transformed complex CSI while conditioning its representation on the acquisition configuration. Contrastive objectives further align CSI samples with instance descriptions and coarse propagation prototypes.

Target-specific branches recover heterogeneous physical attributes using delay-domain features for temporal quantities, raw-amplitude statistics for power-related attributes, circular representations for angles, and coarse-to-fine estimation for continuous targets. Before verbalization, a lightweight relational correction module removes impossible values and reconciles known dependencies among the predicted attributes. A deterministic verbalizer then converts the corrected record into natural language while suppressing inapplicable fields. Consequently, every reported category and numerical value can be traced to an explicit physical prediction rather than being freely generated by a language model.

We evaluate CSI2Text-PGMT on a LoS-balanced dataset

containing 50,000 samples and compare it with conventional neural baselines and four direct CSI-to-text decoders based on Qwen3-1.7B, Qwen3.5-2B, InternLM3-8B, and DeepSeek-R1-Qwen3-8B. The proposed model achieves a first-path-delay mean absolute error of 21.91 ns and produces channel reports with 96.43% composite factuality. Although the direct language decoders generate fluent and nearly complete records, their delay-slot tolerance accuracy ranges from only 17.81% to 28.33%, compared with 96.05% for our structured pipeline. The framework also maintains high description factuality under unseen frequency and array configurations, demonstrating that its advantage is not limited to a single acquisition setting. Further robustness and stratified analyses identify NLoS first-path inference as the main remaining failure mode, revealing a concrete direction for future multipath-aware modeling and uncertainty calibration.

To further assess practical utility, we add five downstream wireless-control tasks. The learned interface gives a 34.15 m mean localization error with 49.02% of links within 10 m, reaches 70.92% accuracy for adaptive cyclic-prefix selection, and estimates link quality with a 2.1098 dB power MAE while retaining 98.92% outage accuracy. For resource-management decisions, it achieves 99.05% MCS accuracy with only 0.53% BLER violations and 99.90% oracle goodput, and obtains 87.84% Top-1 and 96.52% Top-3 accuracy for 64-beam transmit-beam search with only 1.33 dB mean gain loss.

Our **main contributions** are summarized as follows:

i. We formulate CSI-to-language conversion as physics-grounded structured inference followed by faithful verbalization, separating physical correctness from linguistic plausibility.
ii. We propose CSI2TEXT-PGMT, a configuration-aware multitask architecture that combines semantic alignment, target-specific physical branches, relational correction, and deterministic verbalization to produce traceable channel reports.
iii. We demonstrate that structured prediction grounds channel descriptions substantially more accurately than direct learned decoding, and use cross-configuration, robustness, and stratified evaluations to identify NLoS multipath inference as the dominant remaining challenge.

The code is available at https://github.com/yanrunhui/CSI_clip_model.

II. Preliminaries

a) Multipath CSI and delay observability.: After retaining one receive element, the frequency-domain CSI is represented by a complex matrix $\mathbf{H}$ across transmit antennas and subcarriers. Under a multipath model,

$$H_{m,k} = \sum_{p=1}^{P} \alpha_p a_m(\phi_p) e^{-j2\pi f_k \tau_p} + n_{m,k},$$

Here, $m \in \{1, \ldots, N_{\mathrm{tx}}\}$, $k \in \{1, \ldots, N_f\}$, and $p \in \{1, \ldots, P\}$ index the transmit antenna element, subcarrier, and propagation path, respectively, where P is the number of retained paths. Moreover, α_p, τ_p, and ϕ_p denote the complex gain, delay, and transmit-side direction of path p; f_k is the k-th subcarrier frequency; $a_m(\phi_p)$ is the corresponding transmit-array response at antenna m; and $n_{m,k}$ is complex measurement noise. Thus, spatial information is encoded across the transmit-array dimension, while delay is encoded by phase variation across frequency. For a dominant path, adjacent subcarriers satisfy

$$H_{m,k+1} H_{m,k}^{*} \propto e^{-j2\pi \Delta f \tau}.$$

The superscript $(\cdot)^*$ denotes complex conjugation, and $\Delta f = f_{k+1} - f_k$ is the adjacent-subcarrier spacing. This relation shows that propagation delay can be inferred from frequency-dependent phase. Its resolution depends on the observed bandwidth and subcarrier spacing, motivating the use of acquisition configuration in the model. Under multipath propagation, overlapping components make a single closed-form estimate insufficient.

b) Beam-delay representation and AoA observability.: We expose the spatial and temporal structures using the beam-delay power representation

$$\mathbf{P}_{\mathrm{bd}} = \left| \mathrm{IFFT}_f(\mathrm{FFT}_{\mathrm{array}}(\mathbf{H})) \right|^2 .$$

Here, $\mathbf{P}_{\mathrm{bd}}$ denotes the beam-delay power representation obtained by applying a spatial FFT over the transmit array, an IFFT over frequency, and an element-wise magnitude square. A propagation path ideally produces concentrated energy in this domain, motivating our beam tokens and delay-specific branches. The retained spatial aperture is on the transmit side, whereas the supervised target is receiver-side AoA. Because only one receive element is retained, classical receive-array estimators cannot directly recover this angle. The model therefore learns receiver-side AoA from the statistical relationship between transmit-side beam-frequency CSI, multipath structure, and ray-traced geometry. Accordingly, AoA is treated as a supervised cross-domain prediction target rather than a directly measured beam direction.

III. Method

CrossConfig-CSI converts heterogeneous CSI $\mathbf{H}$ and acquisition configuration $\mathbf{c}$ into a structured record and an auditable report:

$$\hat{\mathbf{y}} = f_\Theta(\mathbf{H}, \mathbf{c}), \qquad \hat{s} = V(\mathcal{C}(\hat{\mathbf{y}})),$$

where $\mathcal{C}$ corrects known relations and V deterministically verbalizes the resulting record.

Section III-A unifies heterogeneous CSI through phase alignment, frequency resampling, and beamspace tokenization. Section III-B learns a configuration-conditioned shared representation with training-only text and prototype alignment. Section III-C predicts temporal, angular, power, and multipath fields with structure-matched branches. Section III-D corrects known relations and deterministically verbalizes the record, completing a traceable CSI-to-report pipeline.

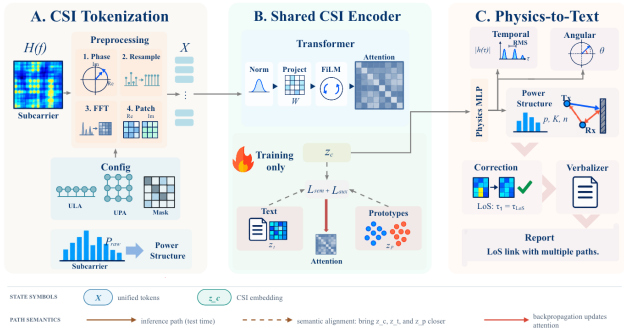


Fig. 1: Overview of CSI2TEXT-PGMT. Stage A phase-aligns, frequency-resamples, and transforms the heterogeneous CSI into unified beam-frequency tokens while retaining a raw-power bypass. Stage B uses configuration conditioning and a shared Transformer encoder; text and prototype embeddings provide semantic supervision only during training. Stage C maps the shared CSI embedding to temporal, angular, and power/structure predictions, applies relational correction, and deterministically verbalizes the corrected physical record. Solid arrows denote the test-time inference path, dashed arrows denote training-only semantic alignment, and the red arrow shows the backpropagation update to attention.

A. Unified CSI Tokenization

Consider frequency-domain CSI

$$\mathbf{H} \in \mathbb{C}^{N_{\mathrm{rx}} \times N_{\mathrm{tx}} \times N_f}.$$

Because its dimensions vary with array geometry and frequency sampling, we transform each observation into

$$(\mathbf{X}, \mathbf{M}, \mathbf{B}) = \mathcal{T}(\mathbf{H}, \mathbf{c}),$$

where $\mathbf{X}$ is the unified token tensor, $\mathbf{M}$ its validity mask, and $\mathbf{B}$ the normalized beam positions.

We remove arbitrary global phase using a reference coefficient,

$$\mathbf{H}^{(a)} = \mathbf{H} \exp(-j \angle H_{\mathrm{ref}}).$$

and resample through the delay domain to a fixed $N_f^\star$ before the beamspace transform

$$\mathbf{H}_{\mathrm{beam}} = \begin{cases} \mathrm{FFT}_{\mathrm{ant}}(\mathbf{H}^{(a)}), & \text{ULA}, \\ \mathrm{FFT2}_{\mathrm{row,col}}(\mathbf{H}^{(a)}), & \text{UPA}. \end{cases}$$

Adjacent ULA beams or local UPA beam patches form tokens by concatenating real and imaginary components; $\mathbf{M}$ excludes padded boundaries.

B. Configuration-Aware Representation Learning

The valid elements of each example are standardized before entering the shared encoder:

$$\widetilde{\mathbf{X}} = \frac{\mathbf{X} - \mu_{\mathbf{X}}}{\sigma_{\mathbf{X}} + \epsilon},$$

where the mean and standard deviation are computed only over tokens selected by $\mathbf{M}$. This removes link-dependent scale variation.

Each token is projected with linear, depthwise-pointwise convolutional, and adaptive-pooling layers. Subcarrier spacing modulates the result through

$$\mathrm{FiLM}(\mathbf{u}; \Delta f) = (1 + \boldsymbol{\gamma}(\Delta f)) \odot \mathbf{u} + \boldsymbol{\beta}(\Delta f).$$

Beam-position and configuration encodings are then added. A shared Transformer fuses a learned [CLS] token with the masked token mean:

$$\mathbf{z}_c = \mathrm{Norm}\left[\mathcal{P}_c\left(\mathbf{Z}_{\mathrm{cls}} + \frac{\sum_k M_k \mathbf{Z}_k}{\sum_k M_k}\right)\right].$$

a) Training-only semantic alignment.: During training, a text encoder maps description s_i to $\mathbf{z}_t^i$, and a prototype bank maps coarse semantic key k_i to $\mathbf{z}_p^i$:

$$\mathbf{z}_t^i = f_{\mathrm{text}}(s_i), \qquad \mathbf{z}_p^i = \mathbf{P}(k_i).$$

The key summarizes environment, LoS/NLoS state, path richness, and coarse physical regimes. We use symmetric contrastive alignment:

$$\mathcal{L}_{\mathrm{align}} = \mathcal{L}_{\mathrm{NCE}}(\mathbf{z}_c, \mathbf{z}_t) + \mathcal{L}_{\mathrm{NCE}}(\mathbf{z}_c, \mathbf{z}_p) + \mathcal{L}_{\mathrm{NCE}}(\mathbf{z}_t, \mathbf{z}_p).$$

Semantic and balanced K-factor-bin classifiers add supervision; all text/prototype branches are removed at inference.

C. Physics-Aware Structured Prediction

A residual MLP first maps the shared embedding to a physics-oriented representation:

$$\mathbf{z}_{\mathrm{phy}} = \mathbf{z}_c + \alpha g_{\mathrm{shared}}(\mathbf{z}_c).$$

Target-specific branches predict categorical, temporal, angular, power, and multipath fields. The temporal branch combines $\mathbf{z}_{\mathrm{phy}}$ with delay-domain features and uses coarse-bin classification followed by within-bin regression:

$$\hat{y} = l_{\hat{b}} + \hat{r}\left(u_{\hat{b}} - l_{\hat{b}}\right),$$

where $\hat{b}$ is the interval, $\hat{r}$ its normalized position, and $l_{\hat{b}}, u_{\hat{b}}$ its bounds.

The angular branch uses sine/cosine outputs to respect periodicity:

$$\hat{\theta} = \mathrm{atan2}(\hat{s}_\theta, \hat{c}_\theta).$$

a) Raw-scale power bypass.: Because standardization removes absolute amplitude, unstandardized tokens provide a raw-scale feature:

$$\mathbf{r}_{\mathrm{raw}} = g_{\mathrm{raw}}(\mathbf{X}, \mathbf{M}).$$

The power head combines it with the shared physics representation:

$$\hat{p}_1 = g_{\mathrm{power}}([\mathbf{z}_{\mathrm{phy}}; \mathbf{r}_{\mathrm{raw}}]).$$

Let $\mathcal{Q}$ denote the set of physical targets. The overall training objective is

$$\mathcal{L} = \sum_{q \in \mathcal{Q}} \lambda_q \mathcal{L}_q + \lambda_{\mathrm{align}} \mathcal{L}_{\mathrm{align}} + \lambda_{\mathrm{sem}} \mathcal{L}_{\mathrm{sem}} + \lambda_{\mathrm{attr}} \mathcal{L}_{\mathrm{attr}},$$

where $\mathcal{L}_q$ matches the target type; weights and intervals are detailed in the appendix.

D. Relational Correction and Verbalization

Specialized heads may violate simple relations, so a deterministic correction precedes text generation:

$$\widetilde{\mathbf{y}} = \mathcal{C}(\hat{\mathbf{y}}).$$

It clamps invalid negative spreads/counts, enforces at least one path, and equates earliest-path and direct-LoS delay for predicted LoS samples:

$$\widetilde{\tau}_1 = \widetilde{\tau}_{\mathrm{LoS}} = \hat{\tau}_{\mathrm{LoS}}.$$

The correction is non-learned. A deterministic verbalizer then produces

$$\hat{s} = V(\widetilde{\mathbf{y}}).$$

Conditional templates suppress inapplicable fields; every emitted value comes from the corrected record.

During inference, the complete path is

$$(\mathbf{H}, \mathbf{c}) \to \mathbf{X} \to \mathbf{z}_c \to \hat{\mathbf{y}} \to \widetilde{\mathbf{y}} \to \hat{s},$$

with all semantic supervision removed.

IV. Experiments

We organize the evaluation around seven questions. Section IV-A compares physical estimators and general-purpose language decoders, then examines measured inference cost. Section IV-B tests generalization across frequency samplings, antenna arrays, spatial blocks, and unseen maps. Section IV-C studies robustness to complex AWGN, Section IV-D isolates semantic alignment and physical prediction-head components, and Section IV-E analyzes the dominant LoS–NLoS failure modes. Section IV-F evaluates uncertainty-based selective prediction, while Section IV-G validates the delay pipeline on measured SISO data.

A. Main Results: Physical Grounding and CSI-to-Language

1) Comparison with Conventional Physical Estimators: We first compare CSI2Text-PGMT with representative conventional CSI-based physical estimators. The baselines include PDP/IFFT + MLP, which transforms frequency-domain CSI into delay-domain features before regression; Flattened CSI + MLP, which directly vectorizes the complex CSI observation; a Transformer without task-specific branches; a CNN baseline over the beam–frequency input; and a non-learned IFFT–PDP estimator that reads delay and power statistics from the power-delay profile. A CSI encoder with single-task heads is also included to isolate the effect of sharing the encoder while predicting each target independently.

All methods are evaluated on the D2Los_Data split under the same UPA-8×8 setting with 128 CSI frequency points. For each seed, 40,000 training samples are randomly drawn from the source pool and evaluated on 10,000 held-out samples. LoS and NLoS links are balanced in both training and test sets to avoid visibility bias, and learned models are trained with three random seeds. As shown in Figure 1, CSI2Text-PGMT uses the same CSI observation as input but routes the shared representation through physics-aware branches before deterministic report generation.

TABLE I: Main balanced-split physical results. Learned methods report mean ± sample standard deviation over three seeds; the deterministic IFFT–PDP baseline is evaluated once on 10,000 test examples. Lower is better for MAE; higher is better for reflection-bin accuracy. Underlining marks the best result across all methods, including ours. The final row gives the relative change from the best prior method, with direction indicated by color (red: improvement; blue: degradation).

Method	Delay (ns) ↓	Angle (°) ↓	Power (dB) ↓
CSI2Text-PGMT	**21.911±0.303**	**18.372±0.365**	**4.958±0.269**
CSI encoder + single-task heads	51.164±6.248	20.449±0.431	5.074±0.209
PDP/IFFT + MLP	28.750±4.611	40.292±0.159	12.236±0.278
Flattened CSI + MLP	114.371±0.978	32.782±0.413	13.315±1.221
Transformer without branches	145.177±0.003	42.614±0.005	28.091±0.012
CNN baseline	169.671±0.021	53.111±0.005	38.090±0.013
IFFT–PDP (non-learned)	160.262	N/A	6.555
Rel. change vs. best prior	**23.8%**	**10.2%**	**2.3%**

Method	K (dB) ↓	Refl. MAE ↓	Bin acc. ↑
CSI2Text-PGMT	**1.983±0.263**	**0.983±0.005**	**0.7061±0.0039**
CSI encoder + single-task heads	1.521±0.090	1.222±0.072	0.5523±0.0259
PDP/IFFT + MLP	5.002±0.275	1.706±0.032	0.4420±0.0201
Flattened CSI + MLP	5.600±1.013	1.835±0.196	0.4237±0.0999
Transformer without branches	20.762±0.051	3.214±0.021	0.1808±0.0008
CNN baseline	40.852±0.055	10.134±0.011	0.1562±0.1463
IFFT–PDP (non-learned)	20.063*	N/A	N/A
Rel. change vs. best prior	**30.4%**	**19.6%**	**27.8%**

*Dominant-to-residual ray-power-ratio heuristic, not a multi-snapshot Rician estimator. The non-learned baseline additionally obtains path-count MAE 1.6445, delay-spread MAE 36.570 ns, and LoS accuracy 84.72%. Receiver-side angle and reflection outputs are unavailable under the single-receive-element observation.

Table I shows that conventional estimators can fit individual attributes but struggle to provide a complete and consistent propagation record. The non-learned IFFT–PDP pipeline is useful for direct delay-domain cues, yet it cannot output receiver-side angle or reflection fields. Flattened MLP and CNN models are much weaker on delay, angle, power, and multipath quantities because they do not explicitly preserve the coupled beam–frequency evidence. The generic Transformer has higher capacity but still lacks the target-matched temporal, angular, power, and structure branches. By contrast, CSI2Text-PGMT obtains the best delay, angle, power, reflection-MAE, and reflection-bin accuracy, with relative gains of 23.8%, 10.2%, 2.3%, 19.6%, and 27.8%, respectively, over the best prior baseline. The only exception is the scalar K-factor result of the single-task baseline, which indicates that the proposed design is strongest for coupled channel reporting rather than for every isolated scalar target.

2) Comparison with General-Purpose Language Models: We next compare with direct CSI-to-language baselines built on general-purpose LLMs. Qwen3-1.7B and Qwen3.5-2B are compact open language models, InternLM3-8B provides a larger instruction-following backbone, and DeepSeek-R1-Qwen3-8B represents a reasoning-oriented decoder. These models are tested as direct report generators rather than as post-processing modules for the structured predictions of CSI2Text-PGMT.

The experimental route is CSI → CSIEncoder → MLP pro-

jection → continuous prefix embedding → LLM → physical-attribute JSON and textual description. The LLM receives continuous vectors learned from CSI, while the prompt supplies the observation configuration, including array type, bandwidth, and subcarrier spacing, and requires a fixed output schema. During training, the language-model loss is computed only on the target answer; tokens corresponding to the CSI prefix and prompt are masked with -100 and do not contribute to the loss. Depending on the run configuration, the CSIEncoder can be trained, frozen, or initialized from existing weights, while the LLM backbone is frozen and optional LoRA adapters can be trained. The data split follows the same balanced evaluation protocol as above, with the reported decoder comparison using the common 1,000-example cohort.

TABLE II: Structured prediction versus direct general-purpose language decoding on the common 1,000-example cohort. Panel A reports description-level quality; Panel B reports field-level error. Higher is better for accuracy and slot F1; lower is better for hallucination and MAE. Underlining marks the best result across all methods, including ours. Signed changes are computed against the best direct decoder; direction is indicated by color (red: improvement; blue: degradation).

Panel A: description-level quality

Method	Factuality (%) ↑	Numeric acc. (%) ↑	Slot F1 (%) ↑
CSI2TEXT-PGMT	**96.43**	**82.61**	**99.96**
Qwen3-1.7B	92.47	64.15	99.85
Qwen3.5-2B	92.68	65.06	99.86
InternLM3-8B	92.76	65.71	99.84
DeepSeek-R1-Qwen3-8B	83.27	44.72	97.82
Rel. change vs. best prior	**4.0%**	**25.7%**	**0.1%**

Method	Halluc. (%) ↓	Delay acc. (%) ↑	Delay MAE (ns) ↓
CSI2TEXT-PGMT	**0.074**	**96.05**	**21.91**
Qwen3-1.7B	0.14	22.60	177.66
Qwen3.5-2B	0.14	28.33	147.39
InternLM3-8B	0.16	26.45	165.56
DeepSeek-R1-Qwen3-8B	4.00	17.81	232.66
Rel. change vs. best prior	**47.1%**	**239.0%**	**85.1%**

Panel B: field-level MAE

Method	First delay (ns) ↓	LoS delay (ns) ↓	Spread (ns) ↓	First angle (°) ↓	LoS angle (°) ↓
CSI2TEXT-PGMT	**24.75±1.04**	**8.83±0.55**	**11.30±0.26**	**16.61±0.56**	**1.33±0.17**
Qwen3-1.7B	177.66	146.54	47.03	42.84	28.71
Qwen3.5-2B	147.39	89.09	44.48	40.60	27.66
InternLM3-8B	165.56	127.86	46.57	40.22	29.07
DeepSeek-R1-Qwen3-8B	232.66	175.22	49.64	59.16	36.05
Rel. change vs. best direct LLM	**83.2%**	**90.1%**	**74.6%**	**58.7%**	**95.2%**

Method	Az. spread (°) ↓	First power (dB) ↓	K-factor (dB) ↓	Refl. MAE ↓	Path MAE ↓
CSI2TEXT-PGMT	**9.72±0.47**	**4.34±0.27**	**2.15±0.04**	**1.032±0.019**	**0.086±0.005**
Qwen3-1.7B	8.16	6.17	2.559	1.409	0.157
Qwen3.5-2B	6.76	5.80	2.360	1.443	0.160
InternLM3-8B	8.10	5.96	2.485	1.494	0.163
DeepSeek-R1-Qwen3-8B	53.52	27.77	16.418	3.618	0.578
Rel. change vs. best direct LLM	**43.8%**	**25.2%**	**8.9%**	**26.8%**	**45.2%**

Table II shows that direct LLM decoders can learn the output schema and produce fluent-looking channel reports, but their numerical grounding is much weaker. The best direct decoder reaches only 65.71% numerical accuracy, whereas CSI2TEXT-PGMT reaches 82.61%. The gap is more pronounced for physical fields: CSI2TEXT-PGMT reduces first-delay MAE by 83.2%, LoS-delay MAE by 90.1%, first-angle MAE by 58.7%, and LoS-angle MAE by 95.2% relative to the best direct LLM. It also lowers first-power, K-factor, reflection, and path-count errors. These results indicate that the main advantage of CSI2TEXT-PGMT is not text fluency alone, but the separation of physical inference, relation correction, and deterministic verbalization.

3) Measured Inference Cost: A deployable channel-reporting system must be accurate without imposing the latency and memory footprint of autoregressive generation. We therefore conduct a controlled inference-cost measurement: all methods are evaluated on the same CSI test samples, the same GPU, and a fixed warmup/repeat protocol. We record end-to-end latency, CUDA time, throughput, peak allocated VRAM, parameter count, and generated-token statistics, and then merge these measurements with the corresponding physical and text-quality metrics. We therefore measure end-to-end latency on 200 common examples and peak allocated VRAM under explicit precision settings; the quality results use the separate 1,000-example decoder cohort.

TABLE III: Measured end-to-end inference cost. Conventional models use FP32, whereas direct language decoders use 4-bit NF4 weights with BF16 computation. Lower is better. Underlining marks the best result across all methods, including ours. The final row compares CSI2TEXT-PGMT with the most efficient direct language decoder for each cost measure.

Method	Median (ms) ↓	P95 (ms) ↓	VRAM (GB) ↓	Params ↓
CSI2TEXT-PGMT-COST-BENCHMARK	**13.58**	**14.41**	**0.13**	**32.1M**
Single-task	13.47	13.78	0.23	58.9M
PDP/IFFT+MLP	3.76	3.85	0.02	3.3M
Flattened MLP	1.63	1.82	0.33	86.5M
Transformer	7.44	7.58	0.11	28.1M
CNN	3.44	3.49	0.04	6.5M
Qwen3-1.7B	12446	15391	1.53	1.07B
Qwen3.5-2B	11656	14450	2.03	1.41B
DeepSeek-8B	16905	33112	6.18	4.81B
InternLM3-8B	23046	27953	6.18	5.03B
Rel. change vs. best direct LLM	**99.9%**	**99.9%**	**91.5%**	**97.0%**

Fig. 2: Quality–cost Pareto comparison. The left panel relates description factuality to median inference latency, and the right panel relates numerical tolerance accuracy to latency. Bubble area encodes peak allocated VRAM. The structured model occupies the high-quality, low-latency corner in both views.

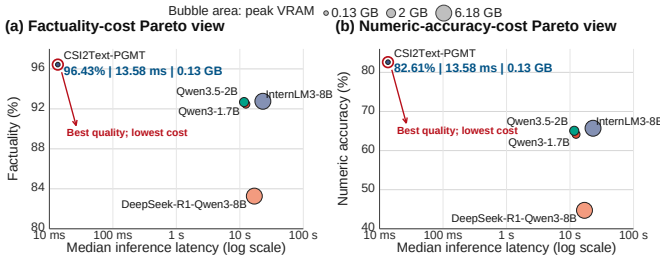


Table III and Figure 2 show that direct language decoding pays a large systems cost in addition to its weaker numerical grounding. The direct decoders operate on a seconds-per-report scale and allocate far more memory, whereas CSI2TEXT-PGMT remains in the millisecond regime with a compact physical predictor. Conventional regressors can be slightly faster, but they do not approach the complete quality profile in Table I. The proposed pipeline therefore occupies the favorable

accuracy–cost corner: it preserves structured factuality without requiring autoregressive generation for every report.

This distinction matters for radio-network operation, where many candidate links or beams may need to be evaluated within one scheduling interval. The LLM baselines must keep large backbones resident and generate output tokens even for a fixed report schema. In contrast, CSI2TEXT-PGMT predicts the structured record in one feed-forward pass and then uses deterministic verbalization, retaining physically auditable reports while staying close to the cost profile of compact neural estimators.

B. Generalization

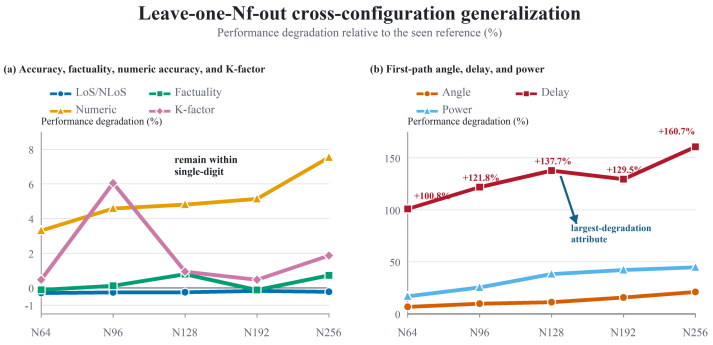


Fig. 3: Leave-one-N_f-out generalization. The single-column two-panel line plot reports performance degradation relative to the seen-$N_f = 128$ reference across held-out frequency samplings. Panel (a) summarizes accuracy, factuality, numeric accuracy, and K-factor trends, whereas panel (b) highlights first-path angle, delay, and power. Positive values indicate worse held-out performance.

1) Frequency-Configuration Generalization: For each column of Figure 3, the complete model is trained on the other four values in $\{64, 96, 128, 192, 256\}$ and then evaluated on the excluded sampling configuration. Visibility classification and report factuality remain stable across the held-out settings, while fine temporal localization is the most sensitive quantity and power or angle exhibit smaller configuration-dependent changes. The result shows that the configuration-aware representation transfers semantic channel structure more reliably than it transfers exact delay resolution. An assisted setting with paired observations from two numerologies is examined separately as a complementary multi-view extension.

2) Cross-Array and Directional Transfer: We train a mixed-array model with UPA-8×8 and ULA-64 observations, then evaluate it on those seen geometries and on aligned, held-out UPA-4×4, UPA-16×4, and ULA-32 groups without geometry-specific fine-tuning. A separate pair of experiments reverses the ULA transfer direction, comparing larger-to-smaller and smaller-to-larger apertures while keeping the evaluation groups disjoint from training.

TABLE IV: Cross-array and directional ULA transfer over three seeds. UPA-8×8 and ULA-64 are seen during mixed-array training. Lower is better for MAE; higher is better for factuality. Panel B also reports the direction-aware relative change from the seen setting to the unseen setting.

Panel A: cross-array transfer

Array	Status	LoS angle ↓	NLoS angle ↓	First delay ↓	Factuality ↑
UPA-8×8	Seen	6.106±1.966	42.591±0.377	53.854±6.213	94.316±0.096%
ULA-64	Seen	14.566±2.623	42.710±0.562	47.449±2.121	94.178±0.167%
UPA-4×4	Unseen	10.257±3.013	45.859±0.110	44.082±0.786	93.556±0.092%
UPA-16×4	Unseen	9.446±0.913	43.237±0.114	54.828±5.982	93.639±0.136%
ULA-32	Unseen	11.345±0.566	42.919±0.400	43.399±1.029	94.449±0.193%

Panel B: directional ULA transfer

Direction	Status	LoS angle ↓	NLoS angle ↓	First delay ↓	Factuality ↑
64 → 32	Seen 64	14.566±2.623	45.613±0.322	47.449±2.121	94.178±0.167%
	Unseen 32	11.345±0.566	41.740±0.462	43.399±1.029	94.449±0.193%
	Unseen vs. seen	**22.1%**	**8.5%**	**8.5%**	**0.3%**
32 → 64	Seen 32	10.730±0.785	42.982±0.112	41.619±0.954	95.092±0.096%
	Unseen 64	17.223±3.331	48.592±0.628	49.350±3.524	94.222±0.073%
	Unseen vs. seen	**60.5%**	**13.1%**	**18.6%**	**0.9%**

Table IV shows that factual report generation remains stable on unseen array shapes and sizes, while the persistent NLoS-angle error is shared across geometries rather than caused by one specific array. Directional transfer is asymmetric: learning from the larger aperture transfers cleanly to the smaller one, whereas extrapolating from a smaller to a larger aperture produces a clearer loss in angular and temporal precision. The model therefore generalizes across array configurations, but the training aperture still bounds the reliability of geometric extrapolation.

3) Spatial-Block and Map-Disjoint Generalization: To distinguish local interpolation from environment transfer, we compare a random split with $32\,\mathrm{m} \times 32\,\mathrm{m}$ spatial blocks and complete held-out maps. The audited cohorts have zero overlap in maps, map-qualified blocks, and user-equipment positions, so the stricter protocols cannot reuse nearby or identical propagation links from training.

TABLE V: Random, spatial-block, and complete-map-disjoint evaluation. The final column reports the direction-aware performance difference between map-disjoint evaluation and random splitting. Higher is better for accuracy; lower is better for MAE. Red denotes an improvement and blue denotes a degradation. Underlining marks the best result in each row, and boldface highlights the map-disjoint results.

Metric	Random	Spatial block	Map-disjoint	Map-disjoint vs. Random
Semantic accuracy ↑	0.8750	0.8763	**0.8618**	**1.32 pp**
Semantic macro-F1 ↑	0.5960	0.6352	**0.4884**	**10.76 pp**
First delay (ns) ↓	25.8400	31.7410	**38.5178**	**49.1%**
LoS delay (ns) ↓	10.2439	13.4182	**17.5738**	**71.6%**
NLoS delay (ns) ↓	47.4350	50.0639	**59.4618**	**25.4%**
Delay spread (ns) ↓	14.9583	16.5812	**18.6648**	**24.8%**
K (dB) ↓	2.4280	2.7301	**3.2529**	**34.0%**
First power (dB) ↓	4.6009	5.2943	**6.2998**	**36.9%**
First angle (°) ↓	21.8495	23.9274	**21.7308**	**0.5%**
Azimuth spread (°) ↓	7.5784	10.1609	**10.0201**	**32.2%**
Reflection-bin accuracy ↑	0.6599	0.5711	**0.5498**	**11.01 pp**
Path-count MAE ↓	0.1965	0.2321	**0.1878**	**4.4%**

Table V shows that performance remains usable when evaluation moves from random interpolation to spatially or map-disjoint transfer. Delay, power, dispersion, and reflection are more sensitive to unseen environments, whereas first-angle

and path-count prediction remain comparatively stable. The nonuniform change is important: random splitting is optimistic for selected physical quantities, but the retained semantic and structural performance demonstrates meaningful generalization beyond memorized locations.

C. Robustness

We evaluate robustness by perturbing the observed complex CSI with additive white Gaussian noise (AWGN) during both training and testing. For each sample, noise is applied only to the valid, non-padding index set Ω. Given a target signal-to-noise ratio γ in decibels, the corrupted CSI is

$$\widetilde{\mathbf{H}} = \mathbf{H} + \mathbf{N}, \qquad P_H = \frac{1}{|\Omega|}\sum_{u\in\Omega}|H_u|^2, \tag{1}$$
$$P_N = P_H 10^{-\gamma/10}, \quad \Re(N_u), \Im(N_u) \sim \mathcal{N}(0, P_N/2),$$

where $\mathbb{E}|N_u|^2 = P_N$. During noise-augmented training, each batch is independently selected for corruption with $z_b \sim \text{Bernoulli}(0.8)$, and its SNR is sampled as $\gamma_b \sim \mathcal{U}(10, 30)$ dB. After warm-up, the batch objective is

$$\begin{aligned}\mathcal{L}_b = \mathcal{L}_{\text{clean}} &+ 0.2\mathcal{L}_{K,\text{clean}} \\ &+ z_b\big[0.5\mathcal{L}_{\text{noisy}} + 0.1\mathcal{L}_{\tau,\text{cons}} \\ &\qquad + 0.1\mathcal{L}_{K,\text{cons}} + 0.1\mathcal{L}_{K,\text{noisy}}\big],\end{aligned} \tag{2}$$

where the supervised terms are masked Smooth-L1 losses and the consistency terms compare noisy predictions with stop-gradient clean predictions in normalized physical-target space. Testing uses freshly sampled noise at $\gamma \in \{30, 20, 10, 0\}$ dB, together with an uncorrupted reference set, so the evaluation measures generalization to unseen noise realizations rather than reuse of the training perturbations.

For each trained model seed s, every metric is first averaged over three independent test-noise realizations r, and the final mean and sample standard deviation are then computed over three model seeds:

$$\bar{M}_s = \frac{1}{3}\sum_{r=1}^{3} M_{s,r}, \qquad \mu_M = \frac{1}{3}\sum_{s=1}^{3}\bar{M}_s,$$
$$\sigma_M = \sqrt{\frac{1}{2}\sum_{s=1}^{3}(\bar{M}_s - \mu_M)^2}. \tag{3}$$

For a lower-is-better metric, the relative degradation at SNR γ is

$$D_M(\gamma) = \frac{M(\gamma) - M(\text{clean})}{M(\text{clean})} \times 100\%. \tag{4}$$

TABLE VI: Three-seed test-time complex-AWGN robustness. Lower is better for MAE; higher is better for numerical tolerance accuracy.

Metric	Clean	30 dB	20 dB	10 dB	0 dB
First delay (ns) ↓	25.97±0.34	72.70±0.40	86.11±0.96	100.23±1.73	209.29±12.31
LoS delay (ns) ↓	8.72±0.49	20.30±2.71	25.00±2.43	34.10±1.32	231.85±28.52
NLoS delay (ns) ↓	43.22±0.27	125.10±3.05	147.22±2.52	166.36±4.31	186.73±6.54
First angle (°) ↓	18.84±0.40	23.10±0.84	24.29±0.71	25.12±0.79	32.30±1.92
First power (dB) ↓	5.12±0.35	4.88±0.37	4.79±0.37	5.38±0.58	17.69±0.08
K (dB) ↓	2.084±0.124	2.142±0.125	2.612±0.118	5.978±0.582	19.855±0.659
Reflection-path MAE ↓	1.018±0.014	1.180±0.006	1.232±0.011	1.338±0.009	2.384±0.178
Numeric accuracy (%) ↑	74.05±0.57	69.27±0.65	64.91±0.82	56.22±1.01	33.93±2.18

Fig. 4: Test-time complex-AWGN robustness. Panel (a) reports the absolute MAE of first-path, LoS, and NLoS delay; panel (b) reports first-angle, first-power, and K-factor MAE normalized by their clean-condition values, focusing the comparison on physical prediction robustness.

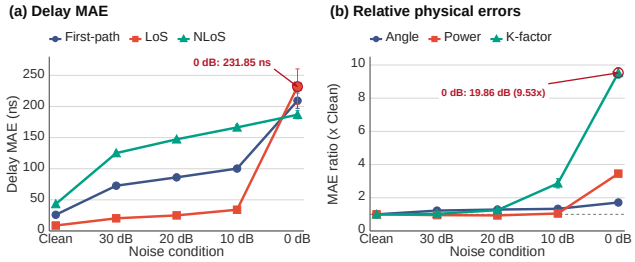


Table VI and Figure 4 show that degradation is monotonic overall but strongly attribute dependent. Delay is affected first because it is encoded by phase variation across frequency. The NLoS delay curve rises much faster than the LoS curve at moderate SNR, which is consistent with the difficulty of locating the earliest indirect component among several overlapping reflections. First-angle error grows more gradually, indicating that coarse spatial structure survives moderate corruption better than fine temporal localization.

Power and K-factor remain close to their clean baselines at moderate noise levels but deteriorate sharply when the SNR approaches 0 dB. The small fluctuation in power error at 20–30 dB lies within the reported seed variation and should not be interpreted as an improvement. Reflection-path error also increases steadily, while numerical tolerance accuracy in Table VI decreases as more fields leave their accepted ranges. Overall, noise augmentation preserves useful physical predictions under moderate corruption, but it cannot recover channel evidence destroyed by severe noise. This attribute-specific behavior supports confidence-aware use: coarse state may remain usable even when a precise delay or power value should be withheld.

D. Ablation

1) Language-Alignment Ablation: Language alignment provides semantic organization for the shared CSI representation before the physical heads produce their structured outputs. The CSI–text objective associates each sample with its complete channel description, whereas text–prototype alignment pulls semantically similar samples toward coarse propagation prototypes such as LoS/NLoS regime and path-richness categories. These language branches are used as training-time regularizers and are removed at inference, so their effect can be isolated without changing deployment cost. Intuitively, the alignment losses do not ask the text branch to generate the final answer; instead, they shape the CSI embedding so that samples with similar propagation semantics occupy nearby regions before the numerical heads are applied. This is useful because several targets are not independent: visibility, path count, reflection strength, and delay spread jointly describe the same propagation regime. A representation that already encodes this coarse regime

gives the downstream heads a narrower and more physically plausible search space.

We construct matched variants that remove CSI–text alignment, text–prototype alignment, or both (the physics-only setting), while keeping the physical heads, target supervision, correction rules, data split, and verbalizer fixed. Table VII therefore measures the contribution of semantic alignment rather than additional inference-time capacity.

TABLE VII: Matched language-alignment ablations over three seeds. Higher is better for classification and bin accuracy; lower is better for MAE. The final row in each block compares the full model with the best ablated variant for each metric (red: improvement; blue: degradation). Underlining marks the best result across all variants, including the full model.

Variant	LoS/NLoS acc. ↑	Delay (ns) ↓	Refl. MAE ↓	Refl. acc. ↑
Full alignment	**0.99572±0.00314**	**25.9495±1.2610**	**1.0644±0.0164**	**0.6866±0.0053**
No CSI–text	0.99297±0.00080	27.3459±0.9163	1.0776±0.0069	0.6579±0.0022
No text–prototype	0.99330±0.00006	28.3746±0.2799	1.0925±0.0100	0.6521±0.0092
Physics only	0.98313±0.00071	27.6446±0.8540	1.1170±0.0135	0.6280±0.0062
Rel. change vs. best ablation	**0.2%**	**5.1%**	**1.2%**	**4.4%**

Variant	Path MAE ↓	NLoS power ↓	NLoS angle ↓
Full alignment	**0.2106±0.0102**	**7.3023±0.2223**	**38.3226±0.7280**
No CSI–text	0.2130±0.0144	7.4089±0.1944	39.6833±0.7642
No text–prototype	0.2144±0.0559	7.3636±0.0424	39.6135±1.2067
Physics only	0.3736±0.0551	8.3348±0.5728	39.2445±1.3988
Rel. change vs. best ablation	**1.1%**	**0.8%**	**2.3%**

The aligned model is strongest on categorical and structure-related labels, especially LoS/NLoS discrimination, path count, reflection-path MAE, and reflection-bin accuracy, while also giving smaller but consistent gains for delay and ambiguous NLoS angle or power. This suggests that language prototypes first narrow the coarse propagation regime and path-structure category, after which CSI–text alignment refines the representation with sample-specific attributes. The two alignment terms therefore play complementary roles: prototypes supply global semantic anchors, while instance descriptions preserve the detailed sample-level relations needed for continuous regression. The largest gains on reflection and path-count metrics are consistent with this interpretation, because those targets depend on recognizing the overall multipath structure rather than reading a single isolated scalar from the CSI. Because every variant still receives direct physical labels, the comparison supports language alignment as a structural regularizer for physical supervision rather than a replacement for it.

2) Physical Prediction-Head Ablations: The physical estimator is deliberately factorized into a shared configuration encoder and shared physics token, followed by multiple target-specific encoders and prediction heads for delay, angle, power, and path structure. This multi-encoder, multi-head design allows coupled evidence to be reused while retaining the inductive bias appropriate to each physical quantity. We remove one component or auxiliary constraint at a time and evaluate only the labels most directly related to that intervention. Table VIII combines the angle, structural, and temporal/power studies.

TABLE VIII: Target-specific physical prediction-head ablations over three seeds. Each panel retains only labels related to the removed component; lower is better unless indicated otherwise. The final row compares the full model with the best ablated variant for each metric (red: improvement; blue: degradation). Underlining marks the best result across all variants, including the full model.

Panel A: angle encoders

Variant	LoS angle ↓	First-angle LoS ↓	First-angle NLoS ↓
Full	**1.3564±0.1419**	**1.6798±0.0434**	**35.2717±0.7477**
No LoS-angle encoder	16.3092±0.2847	5.9300±0.0964	39.8998±1.0083
No first-angle encoder	1.8599±0.2485	2.3328±0.0427	43.5819±0.0074
Rel. change vs. best ablation	**27.1%**	**28.0%**	**11.6%**

Panel B: structural objectives

Variant	Reflection MAE ↓	Reflection acc. ↑	Path-count MAE ↓
Full	**0.9833±0.0054**	**0.7061±0.0039**	**0.1539±0.0079**
No reflection auxiliary loss	8.2848±0.0138	0.1741±0.0028	0.1972±0.0185
No LoS consistency	1.0765±0.1479	0.6548±0.4725	0.2228±0.0088
Rel. change vs. best ablation	**8.7%**	**7.8%**	**22.0%**

Panel C: shared, delay, and power components

Variant	Delay ↓	LoS delay ↓	NLoS delay ↓
Full	**21.9112±0.3031**	**8.1737±0.1556**	**35.8198±0.5986**
No shared token	22.6218±0.7071	9.8156±0.6196	36.6103±0.8507
No delay encoder	69.0200±1.4595	15.9316±0.8742	130.8691±2.2043
No power	29.3139±0.7274	11.0429±1.0211	47.7919±0.2816
Rel. change vs. best ablation	**3.1%**	**16.7%**	**2.2%**

Panel D: spread, K, and power components

Variant	Spread ↓	K ↓	Power ↓
Full	**11.1419±0.1068**	**1.8829±0.2633**	**4.9575±0.2686**
No shared token	11.1766±0.0753	1.7426±0.0419	4.5792±0.0306
No delay encoder	29.6388±0.8355	2.5399±0.1149	5.7443±0.2547
No power	14.1139±0.3321	2.2289±0.1659	4.8605±0.7743
Rel. change vs. best ablation	**0.3%**	**8.1%**	**8.3%**

The effects are localized to the labels matched to each removed component. Without the LoS-angle encoder, LoS-angle estimation deteriorates sharply; without the first-angle encoder, the largest penalty appears on NLoS first-angle prediction. Removing the reflection auxiliary objective produces the most pronounced structural failure, simultaneously worsening reflection regression, reflection-bin classification, and path-count estimation. The LoS-consistency term also supports these coupled structural labels.

The delay encoder is indispensable for temporal estimation: removing it degrades first-path delay, especially in NLoS, and also harms delay spread. Removing the raw-power branch affects not only absolute power evidence but also delay and K-factor, confirming that scale information is shared across the structured record. The shared token shows a smaller, mixed per-label effect, as expected under multitask competition, but generally improves the coherence of the coupled outputs.

Taken together, the panels show why neither a single undifferentiated head nor fully independent regressors are sufficient. Specialized heads provide target-matched representations, while shared encoding transfers propagation-regime, path-prominence, and absolute-scale evidence across related quantities. An occasional isolated metric improvement after removing a component does not overturn the joint result; it reflects reduced multitask competition at the cost of other dependent labels. The full multi-encoder, multi-head architecture is therefore necessary for a consistently accurate and physically coherent channel record.

E. Failure Modes and Error Analysis

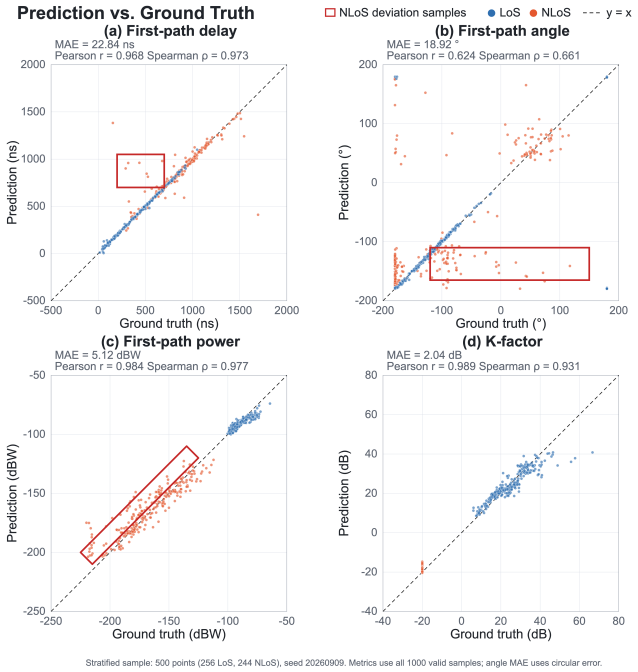


Fig. 6: Per-sample predictions against ground truth for four principal continuous attributes. The display contains a LoS/NLoS-stratified sample of 500 points, while the reported statistics use all 1,000 valid examples. Delay, power, and K-factor closely follow the identity line; first-angle predictions show substantially larger NLoS dispersion. Red boxes identify representative regions containing deviated NLoS samples. Angle error is circular.

Figure 6 first reveals that the dominant failures are concentrated in NLoS samples rather than distributed uniformly across the test set. Delay, power, and K-factor remain strongly ordered overall: their Pearson correlations are 0.968, 0.984, and 0.989, respectively, and most LoS points lie close to the identity line. The largest delay outliers, the compressed high-K region, and the broader power deviations are nevertheless predominantly NLoS cases. First-angle prediction is the clearest failure mode. Its Pearson and Spearman correlations fall to 0.624 and 0.661, and the NLoS samples form broad off-diagonal groups even after circular angle error is used. The figure therefore indicates that the remaining error is principally a propagation-regime and observability problem, not a uniform calibration failure of the model.

Figure 5 quantifies the regime gap exposed by Figure 6. Relative to LoS, NLoS increases first-delay, first-angle, and first-power MAE by $4.38\times$, $21.0\times$, and $2.34\times$, respectively.

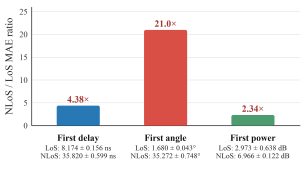


Fig. 5: Propagation-regime error gap. Bars report the NLoS-to-LoS MAE ratio; labels retain the underlying LoS and NLoS mean ± standard deviation.

Consequently, one aggregate MAE can conceal the fact that the balanced test set combines a comparatively easy direct-path regime with a much harder indirect-path regime. The particularly large angle ratio also explains why satisfactory average performance does not imply that every reported NLoS field is equally reliable.

The gap mainly follows from observability. In LoS, the earliest component is tied to a direct path with stable phase, spatial signature, and power ordering. In NLoS, the first retained component is a reflection, and several reflected paths can have similar delays and powers, so small representation errors may change which path is treated as dominant. The problem is amplified because the input keeps only one receive element: receiver-side angle must be inferred indirectly from transmit-side beam–frequency structure and statistical geometry correlations rather than resolved by a receive array. Richer receive-side views, paired numerologies, geometry-aware priors, or NLoS-focused training could reduce this ambiguity; when they are unavailable, high-risk NLoS cases should be deferred or reviewed rather than forced into an automatic report, motivating the selective-prediction mechanism below.

F. Reliability-Aware Selective Prediction

Reliable channel reporting requires more than a low average error: the system must also recognize when an individual prediction should not be trusted. This is especially important for the NLoS tail identified in Section IV-E, where a small number of large delay or angle errors can affect downstream beam management, localization, and channel interpretation even when the aggregate metric remains acceptable. Reliability-aware selective prediction converts this requirement into an explicit coverage–risk trade-off. Low-uncertainty reports are emitted automatically, while uncertain samples are deferred for an additional observation, an alternative frequency configuration, or human review. In this setting, deferral is not treated as a failure of the predictor. It is an operational decision: the system should provide a report when the physical evidence is stable, and it should withhold or verify a report when the model ensemble disagrees about the underlying channel attributes. This is especially relevant for autonomous wireless controllers, where a single erroneous NLoS angle or delay estimate can lead to a wrong beam, a poor localization update, or an overconfident channel summary.

We estimate uncertainty from disagreement among $S = 3$ independently trained models. For sample i, the ensemble prediction and its sample standard deviation are

$$\bar{y}_i = \frac{1}{S}\sum_{s=1}^{S} \hat{y}_i^{(s)}, \qquad u_i = \sqrt{\frac{1}{S-1}\sum_{s=1}^{S}\left(\hat{y}_i^{(s)} - \bar{y}_i\right)^2}, \tag{5}$$

where u_i is the sample-level uncertainty score. The realized absolute error and full-coverage MAE are

$$e_i = |y_i - \bar{y}_i|, \qquad \mathrm{MAE}_{\mathrm{full}} = \frac{1}{N}\sum_{i=1}^{N} e_i. \tag{6}$$

Thus, $\bar{y}_i$ is the value that would be reported if no deferral were used, and u_i measures how sensitive that report is to the random seed and learned representation. A high u_i does not prove that the prediction is wrong, but it indicates that the sample lies in a region where the available CSI evidence supports several plausible physical interpretations. After sorting the samples so that $u_{\pi_1} \le \cdots \le u_{\pi_N}$, coverage c retains the $K(c)$ least-uncertain samples. Their selective risk is

$$K(c) = \mathrm{round}(cN), \qquad \mathcal{S}_c = \{\pi_1, \ldots, \pi_{K(c)}\},$$
$$R_{\mathrm{unc}}(c) = \frac{1}{K(c)} \sum_{i \in \mathcal{S}_c} e_i. \tag{7}$$

We compare this ranking with random rejection and an oracle that retains the samples with the smallest realized errors:

$$\bar{R}_{\mathrm{rand}}(c) = \frac{1}{B} \sum_{b=1}^{B} \frac{1}{K(c)} \sum_{i \in \mathcal{S}_c^{(b,\mathrm{rand})}} e_i,$$
$$R_{\mathrm{oracle}}(c) = \frac{1}{K(c)} \sum_{i \in \mathcal{O}_c} e_i. \tag{8}$$

The random curve answers whether the apparent improvement is merely caused by evaluating fewer samples, whereas the oracle gives the best possible ranking after the true errors are known. The distance between the uncertainty curve and these two references therefore separates practical risk awareness from unavoidable ranking imperfection. To test whether predicted uncertainty tracks actual error across quantities with different units, we order each physical metric by its own uncertainty and form percentile bins $\mathcal{B}_{m,j}$. The bin-wise error for metric j is normalized by that metric's full-coverage MAE, and Panel (b) reports the mean over the four principal continuous metrics:

$$\widetilde{e}_{m,j} = 100 \frac{\frac{1}{|\mathcal{B}_{m,j}|} \sum_{i \in \mathcal{B}_{m,j}} e_{i,j}}{\mathrm{MAE}_{\mathrm{full}}^{(j)}}, \qquad \bar{e}_m^{\mathrm{all}} = \frac{1}{4} \sum_{j=1}^{4} \widetilde{e}_{m,j}. \tag{9}$$

We use the 0–50, 50–60, …, 90–100 percentile ranges; moving right therefore means that the model considers the samples progressively less reliable. A value of 100% equals the average full-coverage error for the corresponding metric before cross-metric averaging. Finally, for a rejection fraction $q = 1 - c$, the retained-set error and relative improvement are

$$\mathrm{MAE}_{\mathrm{unc}}(q) = R_{\mathrm{unc}}(1-q),$$
$$\Delta_{\mathrm{MAE}}(q) = \frac{\mathrm{MAE}_{\mathrm{full}} - \mathrm{MAE}_{\mathrm{unc}}(q)}{\mathrm{MAE}_{\mathrm{full}}}. \tag{10}$$

Thus, $R_{\mathrm{unc}}(1) = \mathrm{MAE}_{\mathrm{full}}$, and rejection changes only which reports are emitted automatically; it does not retrain or alter the underlying predictor.

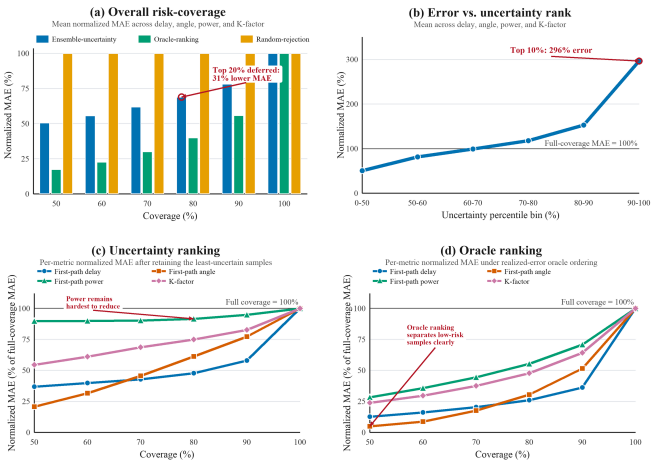


Fig. 7: Reliability-aware selective-prediction diagnostics. (a) Cross-metric normalized risk–coverage comparison at discrete coverage levels for ensemble-uncertainty ranking, oracle ranking, and random rejection. (b) Cross-metric normalized MAE versus uncertainty-percentile bin; 100% denotes the full-coverage MAE. (c) Per-metric normalized MAE for first-path delay, angle, power, and K-factor under uncertainty ranking. (d) The same per-metric normalized MAE under oracle ranking. All values are normalized by the corresponding full-coverage MAE, and lower is better.

Figure 7 shows that ensemble disagreement usefully ranks prediction risk across heterogeneous physical quantities. After normalizing each metric by its full-coverage MAE and averaging delay, angle, power, and K-factor, deferring the highest-uncertainty 10% and 20% reduces retained-set error to about 78% and 69% of the full-coverage baseline, while random rejection remains at 100% in expectation. The bin-wise view is consistent: normalized error rises from roughly 51% for the least-uncertain half to 296% for the highest-uncertainty decile. Although the remaining gap to the oracle shows that disagreement is not perfectly calibrated, the improvement over random rejection supports a practical policy: automatically emit low-uncertainty reports and route deferred samples to additional sensing, paired-numerology fusion, or review. Panel (d) further shows that the gain is strongest for delay and angle, remains useful for K-factor, and is weaker for first-path power.

G. Measured SISO Validation

We evaluate the pipeline on the Wi-World RealData BUPT mobile-measurement scene TXH1_RX01. The acquisition settings in Table IX follow the official dataset documentation [21]. The receiver moves slowly along the measurement route, and each MAT file covers approximately 0.1 s of a 1×1 SISO link.

TABLE IX: Measured-data acquisition, sampling hierarchy, and model input. The independent evaluation unit is a MAT file; snapshots from the same MAT share one short-time measurement interval.

Panel A: acquisition configuration

Item	Setting
Scene	BUPT TXH1_RX01; slowly moving receiver
Carrier / bandwidth	3.7 GHz / 200 MHz
CIR sampling	200 MHz; 5 ns delay interval
Sounding sequence	PN11; length 2047; 100 MSample/s
CIR record	4094 taps; adjacent CIR interval 20.47 μs
MAT contents	Approximately 0.1 s; 4885 or 4886 CIRs
Transmitter	LB-20180 horn; 10.206 dBi gain
Receiver	SZ-3004000/P omnidirectional bicone; 0.952 dBi gain
Channel structure	1×1 SISO

Panel B: data hierarchy

Level	Amount / size	Role
Complete route	$\sim$ 2997 MAT / $\sim$ 900 GB	Original data pool
Local medium pool	400 MAT / 114.72 GiB	Complete MAT and context retained
Debug subset	10 MAT / $\sim$ 2.9 GiB	Pipeline verification; contained in the 60 MAT
Formal measured subset	60 MAT / 17.21 GiB	60 independent evaluation units
Model snapshots	$60 \times 32 = 1920$	Correlated within each MAT

The 60 formal MAT files are uniformly selected in time from the 400-file local pool. They span 17:45:09.200 to 17:49:58.300 on May 30, 2026, for a total of 289.1 s; the median interval between selected MAT files is 4.7 s. Each MAT is treated as one approximate measurement location and one independent evaluation unit. The 10 debug MAT files are included in these 60 files and therefore must not be counted as 10 additional locations.

Each MAT contains approximately 4885 consecutive CIRs. We uniformly sample 32 snapshots over its full 0.1 s interval, giving

$$N_{\text{input}} = 60 \text{ MAT} \times 32 \frac{\text{snapshots}}{\text{MAT}} = 1920 \text{ CSI snapshots}. \tag{11}$$

The 1920 model inputs are not 1920 statistically independent measurement points: the 32 snapshots within a MAT share the same short-time environment. Each snapshot is converted to 256 frequency points over 200 MHz, with 781.25 kHz spacing, and represented by real and imaginary channels with shape $(1, 2, 256)$. A complete MAT therefore produces an input tensor of shape $(32, 1, 2, 256)$, with a 1.28 μs unambiguous delay window.

The preprocessing pipeline reads the complex CIR, samples the 32 snapshots, computes their shared mean power-delay profile (PDP), estimates the noise floor, and constructs one shared soft path mask. The masked complex CIR is transformed by FFT, resampled to the 256-point frequency grid, corrected for the fixed system-delay linear phase, and normalized by a per-snapshot common phase before forming the SISO input. The shared mask threshold is

$$P_{\text{th}} = \max(P_{\text{peak}} - 25 \text{ dB}, P_{\text{noise}} + 6 \text{ dB}). \tag{12}$$

Tap weights are zero below the noise floor, one above P_{th}, and smoothly interpolated between these limits. Using the same mask for all 32 snapshots in a MAT prevents weak paths from appearing or disappearing solely because of snapshot-level noise.

The detected first arrival contains both propagation delay and a fixed acquisition-system delay:

$$\tau_{\text{detected}} = \tau_{\text{system}} + \tau_{\text{propagation}}. \tag{13}$$

GNSS positions are successfully matched for 58 of the 60 MAT files. From these, 12 lower-envelope geometric-LoS candidates are selected, and their median gives $\hat{\tau}_{\text{system}} = 6199.06$ ns, equivalent to 1239.81 samples at 200 MHz. Across the candidates, the standard deviation is 37.53 ns, the median absolute deviation is 33.43 ns, and the range is 6161.30–6269.06 ns. We define the reference as the *PDP-derived calibrated first-arrival delay*

$$\widetilde{\tau}_{\text{FA}} = n_{\text{first}} \times 5 \text{ ns} - 6199.06 \text{ ns}. \tag{14}$$

This wording is deliberate: the offset is estimated from automatically selected geometric-LoS candidates rather than from an independent hardware calibration, so $\widetilde{\tau}_{\text{FA}}$ is not strict ground truth.

Only two outputs have measurement-derived references in this experiment: $\widetilde{\tau}_{\text{FA}}$ and RMS delay spread τ_{RMS}, the latter obtained from a classical PDP estimator. Other structured outputs, including absolute path power, physical ray count, interaction count, and receiver-side angle, have no independent reference labels in this measured subset. They may be generated by the model, but they are excluded from quantitative accuracy claims. Table X therefore evaluates only the two supported delay quantities.

System-delay calibration reduces first-arrival-delay MAE from 244.78 to 55.25 ns (77.4%), reduces the absolute bias from 244.78 to 51.22 ns, and raises Pearson correlation from 0.929 to 0.963. The remaining negative bias indicates systematic underestimation, although the high correlation shows that sample-to-sample variation is preserved. Importantly, the residual 55.25 ns is an end-to-end discrepancy against an automatically constructed reference, rather than pure model error. In signed form, it contains model, GNSS, first-path-detection, and system-delay-calibration terms. The median reported RX-position standard deviations are approximately 8.2 m, 11.4 m, and 9.7 m along the latitude, longitude, and height coordinates; as an order-of-magnitude reference, a 10 m range error corresponds to about 33 ns of propagation delay. A weak physical first path may also fall below the PDP detector threshold, causing a later reflection tens of nanoseconds away to be used as the reference. Finally, the 12 calibration candidates themselves have a 33.43 ns MAD and span 6161.30–6269.06 ns. GNSS uncertainty also propagates into this offset estimate, so these terms are not independent and the full 55.25 ns cannot be attributed to the predictor alone. For τ_{RMS}, soft-mask AWGN fine-tuning attains 15.941 ns MAE; AWGN training from scratch gives the best overall result of 13.442 ns MAE, 16.008 ns RMSE, and $r = 0.891$.

TABLE X: Measured SISO delay results on BUPT. Panel A evaluates system-delay calibration for the PDP-derived first-arrival reference; Panel B compares τ_{RMS} training and preprocessing. Lower is better for MAE and RMSE; higher is better for Pearson r.

Panel A: PDP-derived calibrated first arrival $\widetilde{\tau}_{\mathrm{FA}}$

Metric	Soft mask,uncalibrated	Soft mask,calibrated
Reference mean (ns)	758.02	758.02
Prediction mean (ns)	513.25	706.81
MAE (ns) ↓	244.78	**55.25**
Bias (ns)	−244.78	**−51.22**
Pearson r ↑	0.929	0.963

Panel B: RMS delay spread τ_{RMS}

Training	BUPT input	Pred. mean	MAE ↓	RMSE ↓	Bias	Pearson r ↑
Clean	Raw	223.994	177.906	189.028	+177.906	0.410
AWGN fine-tuning	Raw	141.609	95.896	115.063	+95.521	0.423
Clean	Soft mask	95.016	48.929	66.988	+48.929	0.548
AWGN fine-tuning	Soft mask	50.583	15.941	25.976	+4.495	0.563
AWGN from scratch	**Soft mask**	**57.597**	**13.442**	**16.008**	**+11.509**	**0.891**

The measured-data experiment should be interpreted as an in-sample validation of preprocessing, calibration, and delay transfer, not as a leakage-free generalization test. In particular, the 6199.06 ns offset is estimated from candidates drawn from the same 60-MAT evaluation subset. An independent calibration route or direct hardware-delay measurement is needed before claiming out-of-sample absolute-delay accuracy.

H. Transfer to Downstream Wireless Tasks

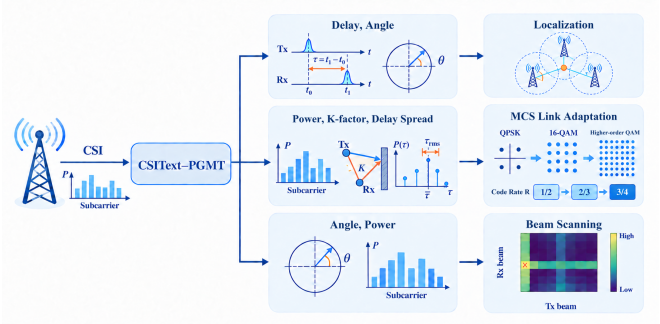


Fig. 8: Representative downstream uses of the learned CSI representation and predicted propagation attributes. The shared representation supports localization, MCS link adaptation, and transmit-beam selection; the quantitative study additionally evaluates adaptive cyclic-prefix selection and link-quality/outage prediction.

We further evaluate five downstream communication tasks to determine whether the learned CSI representation is transferable to practical network decisions rather than being useful only for channel reporting. Figure 8 summarizes the interface: downstream heads receive either CSI features, the pretrained embedding, or the embedding augmented with predicted propagation attributes. Whenever physics features are used, they are model predictions rather than ground-truth test labels. The five tasks cover localization, cyclic-prefix (CP) control, link-quality and outage prediction, MCS adaptation, and transmit-beam selection.

1) Two-Dimensional Localization: For localization, the transmitter and receiver coordinates define the relative-position target $(\Delta x, \Delta y) = (x_{\mathrm{RX}} - x_{\mathrm{TX}}, y_{\mathrm{RX}} - y_{\mathrm{TX}})$. A regression head maps the CSI representation to this two-dimensional offset. This task tests whether the representation retains spatial information that is not explicitly required by the language report. Table XI shows that augmenting a trainable CSI encoder with frozen predicted physics gives the lowest mean and median errors and the largest fractions within 5 and 10 m. The result indicates that the predicted LoS state, delay, and angular structure provide useful geometric cues beyond a generic language backbone.

TABLE XI: Two-dimensional localization on the common test split. Lower is better for mean and median error; higher is better for distance-threshold accuracy. Underlining marks the best result. The final row compares the highlighted model with the best of the three language-model baselines (red: improvement; blue: degradation); threshold accuracies are reported as percentage-point differences.

Method	Mean (m) ↓	Median (m) ↓	≤ 5 m (%) ↑	≤ 10 m (%) ↑
Scratch + frozen physics	**34.15**	**10.67**	**34.04**	**49.02**
Scratch	47.07	20.13	13.36	35.32
Pretrained embedding	65.06	54.46	0.88	4.02
Pretrained embedding + physics	36.91	12.04	29.70	48.20
Qwen3.5-2B, full fine-tuning	100.90	101.17	0.20	0.48
DeepSeek-R1-Qwen3-8B, frozen	100.79	101.09	0.16	0.50
InternLM3-8B, frozen	80.07	78.95	0.86	3.10
Rel. vs. best LLM	**57.3%**	**86.5%**	**33.18 pp**	**45.92 pp**

2) Adaptive Cyclic-Prefix Selection: For each link, offline ray delays and powers define the shortest candidate CP whose protection interval contains at least 99% of the multipath energy. A classifier then selects among the candidate CP durations from the CSI representation and predicted multipath attributes. The objective is to avoid an unsafe short CP while also avoiding excessive overhead from an unnecessarily long CP. Table XII shows a clear tradeoff: pretrained features alone minimize energy outside the chosen CP, whereas adding predicted physics yields the highest classification accuracy and the lowest over-selection rate. Thus, the learned representation supports both reliability-aware and efficiency-aware CP control.

TABLE XII: Adaptive CP selection. Under-selection chooses a CP shorter than the oracle, whereas over-selection chooses a longer CP. Lower outside-energy values are safer. Underlining marks the best result for each directed metric. The final row compares the highlighted model with the best of the three language-model baselines (red: improvement; blue: degradation); predicted CP is descriptive and is therefore omitted from the relative comparison.

Method	Acc. ↑	Under ↓	Over ↓	Pred. CP (ns)	Mean out ↓	P95 out ↓
Pretrained embedding + physics	**70.92%**	**15.92%**	**13.16%**	**215.17**	**4.54%**	**9.35%**
Scratch	51.86%	27.32%	20.82%	189.93	6.51%	55.51%
Pretrained embedding	58.62%	12.22%	29.16%	283.01	2.27%	5.21%
Scratch + frozen physics	67.76%	14.90%	17.34%	221.40	3.57%	5.39%
Qwen3.5-2B, full fine-tuning	51.76%	19.72%	28.52%	290.20	2.80%	10.18%
InternLM3-8B, frozen + CSI	55.58%	21.24%	23.18%	208.43	6.24%	57.31%
DeepSeek-8B, frozen + CSI	58.30%	15.74%	25.96%	247.43	3.27%	11.37%
Rel. vs. best LLM	**21.6%**	**1.1%**	**43.2%**	—	**62.1%**	**8.2%**

3) Link Quality and Outage Prediction: The link-quality task estimates received power and decides whether a link remains operational under the specified noise condition. It therefore tests whether the learned power, LoS, and multipath cues are sufficiently calibrated for reliability-oriented decisions. As shown in Table XIII, the CSI2TEXT-PGMT total-power head gives the lowest power MAE, while all methods exceed 98.9% outage accuracy and F1. The language baselines obtain slightly higher binary scores but incur larger power errors, showing that an accurate outage decision does not by itself guarantee a faithful continuous link-quality estimate.

TABLE XIII: Link-quality and outage prediction. Lower is better for power MAE; higher is better for outage accuracy and F1. Underlining marks the best result. The final row compares the highlighted model with the best of the three language-model baselines (red: improvement; blue: degradation).

Method	Training	Power MAE ↓	Outage acc. ↑	Outage F1 ↑
CSI2TEXT-PGMT	**Total-power head**	**2.1098 dB**	**98.92%**	**98.90%**
DeepSeek-R1-Qwen3-8B	Frozen + adapter/head	3.0165 dB	99.52%	99.51%
InternLM3-8B-Instruct	Frozen + adapter/head	2.8207 dB	99.52%	99.51%
Qwen3.5-2B	Full fine-tuning	2.4603 dB	99.56%	99.55%
Rel. vs. best LLM		**14.2%**	**0.6%**	**0.7%**

4) MCS Link Adaptation: For MCS adaptation, each sample combines the CSI representation, transmit power, and predicted propagation features. Effective-SNR and calibrated BLER constraints define the highest feasible MCS, including an outage class when no MCS satisfies the target. The classifier must therefore maximize spectral efficiency without violating the reliability constraint. Table XIV shows that the scratch CSI2TEXT-PGMT configuration reaches 99.05% accuracy, only 0.53% BLER violations, and 99.90% of oracle goodput. This demonstrates that the structured CSI interface can support near-oracle link adaptation without relying on a large language decoder at inference.

TABLE XIV: MCS link adaptation under calibrated BLER constraints. Lower is better for MCS MAE and BLER violation; higher is better for accuracy, goodput ratio, and outage recall. The final row compares the highlighted model with the best of the three language-model baselines (red: improvement; blue: degradation).

Method	Acc. ↑	MCS MAE ↓	BLER viol. ↓	Goodput ↑	Outage recall ↑
Scratch	**99.05%**	**0.0098**	**0.53%**	**99.90%**	**99.49%**
Pretrained embedding	91.80%	0.1016	3.56%	96.67%	95.94%
Pretrained embedding + physics	92.20%	0.0948	4.40%	97.11%	95.21%
Scratch + frozen physics	98.07%	0.0321	1.38%	99.63%	98.02%
Qwen3.5-2B	95.44%	0.0458	1.22%	98.54%	99.59%
DeepSeek-R1-Qwen3-8B	94.56%	0.0544	4.76%	98.00%	98.29%
InternLM3-8B	94.98%	0.0504	2.46%	98.60%	98.62%
Rel. vs. best LLM	**3.8%**	**78.6%**	**56.6%**	**1.3%**	**0.1%**

5) Adaptive Transmit-Beam Search: The beam task reconstructs the 64 candidate beams of an 8×8 UPA and uses the beam with the largest average received gain as the oracle label. A 64-class head predicts the best transmit-beam index from the full CSI representation. This evaluates whether the representation preserves the directional information required for beam management. Table XV shows that scratch training augmented with frozen predicted physics gives the strongest overall result: 87.84% Top-1 accuracy, 96.52% Top-3 accuracy, and only 1.33 dB mean gain loss. The improvement is largest for NLoS links, where indirect multipath cues are particularly valuable.

TABLE XV: Full-CSI transmit-beam prediction over 64 UPA beams. Lower is better for gain loss; higher is better for all other metrics. Underlining marks the best result. The final row compares the highlighted model with the best of the three language-model baselines (red: improvement; blue: degradation).

Method	Top-1	Top-3	≤ 1 dB	Mean ↓	P90 ↓	LoS	NLoS
Scratch + frozen physics	**87.84%**	**96.52%**	**89.92%**	**1.33**	**1.11**	**98.80%**	**76.88%**
Pretrained embedding	87.10%	95.40%	89.40%	1.57	1.60	98.68%	75.52%
Pretrained embedding + physics	87.02%	95.20%	89.10%	1.62	2.05	98.80%	75.24%
Scratch	86.04%	96.28%	88.38%	1.65	2.75	98.88%	73.20%
Qwen3.5-2B	77.30%	91.10%	80.35%	2.53	10.79	92.50%	62.10%
InternLM3-8B-Instruct	77.60%	90.15%	81.25%	2.51	10.05	92.60%	62.60%
DeepSeek-R1-Qwen3-8B	49.00%	76.15%	51.65%	8.04	25.72	66.50%	31.50%
Rel. vs. best LLM	**13.2%**	**5.9%**	**10.7%**	**47.0%**	**89.0%**	**6.7%**	**22.8%**

Across the five tasks, no single feature choice dominates every secondary metric, but the results consistently show that CSI-native representations and predicted propagation attributes provide a compact and effective interface for network control. In particular, physics assistance is most useful for geometry- and multipath-sensitive decisions, while lightweight CSI heads can match or exceed direct language-model baselines at substantially lower inference cost.

V. Conclusion

We presented CSI-to-language conversion as a grounded inference problem rather than open-ended generation. CSI2TEXT-PGMT first recovers a typed channel state using configuration-aware, target-specific physical representations and then converts the corrected record into an auditable description through deterministic verbalization. The resulting reports are structurally complete and substantially more accurate than direct learned decoders on the underlying numerical quantities. The generalization, split-sensitivity, and observation-noise studies further show that the decisive challenge is not fluent language generation, but the recovery of indirect multipath structure, especially under NLoS propagation. This factorization provides an interpretable interface for wireless-channel analytics while making both successes and failures traceable to explicit physical fields. Future work should extend the framework to measured channels, hardware impairments, calibrated uncertainty, and more expressive multipath-aware objectives.

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APPENDIX A
TARGET CONSTRUCTION AND PHYSICAL SCOPE

This appendix defines the simulator-derived quantities represented by the structured record. These definitions establish annotation semantics and reproducibility; they are not additional learned components.

A. Valid Paths, First Path, and Absolute Power

Let $\mathcal{P}$ be the finite set of paths retained by the D2Los/RayVerse export:

$$\mathcal{P} = \{p \mid P_p^{\mathrm{dBW}} \text{ and } \tau_p \text{ are finite}\}.$$

No additional relative-power threshold is applied. The first path is the earliest retained path,

$$p^\star = \arg\min_{p\in\mathcal{P}} \tau_p, \qquad \tau_1 = \tau_{p^\star}, \qquad p_1^{\mathrm{dBW}} = P_{p^\star}^{\mathrm{dBW}}.$$

Simulator path loss L_p^{dB} is converted to received-path power using the fixed transmit power $P_{\mathrm{tx}} = 23$ dBm:

$$P_p^{\mathrm{dBW}} = P_{\mathrm{tx}}^{\mathrm{dBm}} - L_p^{\mathrm{dB}} - 30.$$

The first-path-power target is therefore an absolute value under this transmit-power assumption, not a per-link relative value. This definition motivates the raw-scale bypass in the model: per-example standardization benefits representation learning but removes the amplitude scale required by the power head.

B. Delay Spread, Angular Spread, and the K-Factor Label

With linear power $P_p = 10^{P_p^{\mathrm{dBW}}/10}$, the delay-spread label is the power-weighted RMS spread,

$$\bar{\tau} = \frac{\sum_p P_p \tau_p}{\sum_p P_p}, \qquad \sigma_\tau = \sqrt{\frac{\sum_p P_p(\tau_p - \bar{\tau})^2}{\sum_p P_p}}.$$

Angular spread uses circular statistics. With $w_p = P_p / \sum_q P_q$,

$$R = \left|\sum_p w_p e^{j\theta_p}\right|, \qquad \sigma_\theta = \frac{180}{\pi}\sqrt{-2\log(\mathrm{clip}(R, 10^{-6}, 1))}.$$

For a LoS link with residual reflected paths, the simulator-derived ray-power-ratio label is

$$K_{\mathrm{dB}} = 10\log_{10}\frac{P_0}{\sum_{p\geq 1} P_p}.$$

NLoS links are assigned -20 dB, and a LoS link with no residual path is assigned 30 dB. This is a derived label based on the retained ray powers, not a multi-snapshot Rician K estimator.

C. Path and Reflection-Interaction Semantics

The supplied CSI dataset contains reflected propagation paths and does not include diffraction or scattering components. Consequently, the stored interaction counts can be interpreted as reflection interactions for this dataset. Let c_p be the stored reflection-interaction count of retained path p. The record distinguishes

$$n_{\mathrm{refl}} = \sum_{p\in\mathcal{P}} \min(c_p, 9), \qquad n_{\mathrm{rpath}} = \sum_{p\in\mathcal{P}} \mathbf{1}[c_p > 0],$$

where n_{refl} is the total encoded reflection-interaction count and n_{rpath} is the number of reflected paths. A single path may contain multiple reflections, so these quantities are related but not identical. Path count is read from the export without a new top-K truncation. An audit of 742,818 links from 20 source files found one to five saved paths per link; five is an empirical ceiling of the supplied exports, not a universal physical limit.

D. Observability Boundaries

The retained spatial aperture is on the transmitter side and only one receive element is supplied. Frequency-dependent phase supports delay inference, and the transmitter array supports transmitter-side spatial structure. Receiver-side AoA is therefore treated as a supervised cross-domain target learned from correlations among transmitter-side beam-frequency CSI, multipath structure, and ray-traced geometry. It must not be interpreted as a directly measured receive-array beam direction. For the same reason, classical receive-array estimators such as MUSIC, ESPRIT, and Bartlett are not commensurate baselines for the receiver-side angle labels in this dataset.

APPENDIX B
IMPLEMENTATION AND EVALUATION DETAILS

A. Encoder, Alignment, and Prediction Heads

ULA inputs use an antenna-axis FFT and group four adjacent beams per token. UPA inputs use a row-column two-dimensional FFT and 2×2 local beam patches. Real and imaginary components are concatenated. The shared encoder combines a linear frequency projection, depthwise-pointwise convolution, adaptive average/max summaries, feature-wise configuration modulation, beam-position embeddings, and carrier/frequency encodings.

The Transformer has six layers, hidden width 384, six attention heads, and a 1,536-dimensional feed-forward block. The fused [CLS] and masked-mean representation is projected to a normalized 256-dimensional CSI embedding. Descriptions use a 300-token vocabulary with maximum length 48. A three-layer, four-head Transformer of width 256 produces the text embedding. Semantic prototype keys combine environment, LoS/NLoS status, path richness, delay and angular regimes, K bin, and reflection statistics; classes with fewer than five training samples are discarded.

For completeness, the alignment loss retains the full symmetric pairwise InfoNCE form used in the initial draft. For two normalized embedding families z_a and z_b in a minibatch of size N,

$$\mathcal{L}_{a,b} = -\frac{1}{2N}\sum_{i=1}^{N}\left[\log\frac{\exp\left((z_a^i)^\top z_b^i/T\right)}{\sum_{j=1}^{N}\exp\left((z_a^i)^\top z_b^j/T\right)} + \log\frac{\exp\left((z_b^i)^\top z_a^i/T\right)}{\sum_{j=1}^{N}\exp\left((z_b^i)^\top z_a^j/T\right)}\right]. \quad (15)$$

The initial temperature is $T = 0.07$. Equation 15 is instantiated for CSI–text, CSI–prototype, and text–prototype pairs. Thus

TABLE XVI: Loss configuration used for the main model.

Group	Term	Weight
Alignment	CSI–text / CSI–prototype / text–prototype	1.0 / 1.0 / 1.0
Representation	semantic / attribute / auxiliary physics	0.5 / 0.1 / 0.005
K factor	strong-bin classifier / within-bin position	0.5 / 0.1
First delay	normalized / raw / fused raw	0.08 / 0.001 / 0.02
First delay	bin / position / consistency / tail underestimate	0.02 / 0.01 / 0.01 / 0.01
Delay spread	raw / bin / within-bin position	0.0005 / 0.03 / 0.008
LoS delay	regression / nonnegative / consistency	0.05 / 0.005 / 0.005
Angles	LoS / first path / NLoS first path	0.05 / 0.02 / 0.01
First power	bin / position; NLoS multiplier	0.03 / 0.1; 2.0
Reflections	classifier / regression; NLoS multiplier	0.02 / 0.01; 1.0

both retrieval directions contribute to each pairwise term rather than treating one embedding family as a fixed one-way target.

Temporal quantities use frequency- and delay-domain features, angular quantities use circular representations, and absolute power combines the standardized representation with the raw-scale bypass. Reflection-interaction count and reflected-path count use separate prediction heads because one path may contain multiple reflection interactions.

B. Loss Weights and Optimization

Table XVI records the main loss configuration. The optional relational training regularizer has weight zero in the reported main runs; record-level correction is applied after prediction.

Models train for 100 epochs with AdamW, batch size 128, initial learning rate 3×10^{-4}, weight decay 10^{-2}, five epochs of linear warm-up, and cosine decay to 10^{-5}. Every epoch traverses all 40,000 main training examples. Retaining the final incomplete minibatch gives 313 minibatches per epoch and 31,300 scheduled optimizer steps. Unless stated otherwise, seeds are 0, 1, and 2, and error bars are sample standard deviations. The balanced main model has 28,938,178 trainable parameters. A run uses one NVIDIA RTX 4090 with 23.6 GB memory, PyTorch 2.5.1 with CUDA 12.1, and no automatic mixed precision; a representative run takes approximately 5.5 hours.

C. Data Cohorts and Split Construction

All cohorts originate from the 400,000-link source pool. Tables XVII and XVIII transcribe the cohort plan used across the main, stress-test, decoder, correction, extension, and cost studies. Cohort-qualified names prevent numerical differences across independently sampled experiments from being interpreted as contradictory estimates on one fixed test set.

For complete map-disjoint evaluation, 78 of 97 source maps form a 321,644-example training parent and 19 maps form a 78,356-example evaluation parent. Balanced subsets are sampled only within their parents: 20,000 LoS and 20,000 NLoS links for training, and 5,000 of each for evaluation. Ordered group IDs, maps, map-qualified spatial blocks, and exact UE positions are checked for empty intersection. The spatial-block protocol has no guard band, so complete map separation remains the stronger scenario-disjoint test.

The 400,000-link source pool contains 53,577 LoS and 346,423 NLoS examples. The exact test-map IDs are

$$\{1, 7, 15, 17, 21, 22, 25, 28, 30, 34, \\ 50, 53, 54, 55, 81, 89, 96, 98, 99\}.$$

all remaining map IDs are training-only. The balanced-subset samplers use seed 23,441 for training and seed 23,442 for evaluation.

Split provenance is verified at four levels. First, the ordered training and evaluation group-ID lists are compared directly. Second, each group is expanded to its map ID, map-qualified spatial-block ID, and exact UE position, and every train–test intersection is required to be empty. Third, the ordered 10,000-example evaluation list stored with each run is compared with the audited evaluation manifest. Fourth, each checkpoint record is bound to the exact training manifest and sampling seed used to fit it. SHA-256 hashes of the complete ordered ID lists, rather than hashes of unordered sets or row counts, are retained in the machine-readable audit. These checks detect three leakage failures that a row-count or class-balance check cannot: (i) cohorts with the same LoS/NLoS balance but shared physical links, (ii) different links sampled from a map that should have been held out in its entirety, and (iii) a checkpoint trained from a manifest different from the one associated with the reported evaluation table.

D. Text Metrics and Tolerance Definitions

Predicted and reference descriptions are parsed into typed slots. Categorical fields use accuracy and macro-F1. Numeric-slot F1 measures whether required values are present. Hallucination is the fraction of predicted numeric slots unsupported by the reference schema. Numerical tolerance accuracy checks whether a present value is within 50 ns for delay, 15° for angle, 5 dB for first-path power, 3 dB for K, and one count for path or reflection fields. Composite factuality is the macro average of categorical accuracies, numeric-slot F1, numerical tolerance accuracy, and one minus hallucination. It is not sentence exact match. Raw MAE remains necessary because the tolerance thresholds are application choices.

The tolerance table complements MAE by exposing the fraction of fields that meet application-scale error bands. It also shows why one aggregate numerical-accuracy value is insufficient: LoS-specific geometry is highly reliable, whereas first-path power and reflection multiplicity remain harder.

E. Baselines and Inference Settings

Physics baselines include flattened CSI with an MLP, a CNN, PDP/IFFT features with an MLP, a Transformer without specialized branches, and independent single-task heads on the same CSI encoder. Direct learned decoders map the CSI encoder representation to a learned soft prefix for Qwen3-1.7B, Qwen3.5-2B, InternLM3-8B, or DeepSeek-R1-Qwen3-8B. These decoders learn to emit the same record schema but do not use the target-specific physical heads. The four decoder results are single trained runs; their missing error bars must not be interpreted as zero variance.

TABLE XVII: Core cohorts and isolation rules.

Cohort	Purpose	Train/test	LoS/NLoS	Isolation and use
SRC-400K	source pool	400K	original	no train/test split; source for subsequent sampling
MAIN-50K-v1	main experiment	40K/10K	20K/20K; 5K/5K	78 training and 19 test maps with zero group-ID overlap; full model, conventional baselines, and architectural ablations
SD-50K	scenario isolation	40K/10K	balanced	map-, spatial-block-, UE-position-, and group-disjoint scenario evaluation

TABLE XVIII: Specialized cohorts. Reflection labels are available unless otherwise noted.

Cohort	Purpose	Train/test	Balance	Isolation or label status
CORR-100K	relational correction	80K/20K	balanced	zero shared links/samples; 97 shared maps
RPATH-100K	reflected-path extension	80K/20K	13,628/86,372	random split; map/link isolation not audited; reflected-path labels reconstructed and verified
DD-100K-v2	decoder training	40K/1K	balanced	zero shared links/samples; 97 shared maps; reconstructed labels
COST-50K-RPATH-v2	cost benchmark	40K/10K	balanced	zero shared links/samples; 97 shared maps
Noise-Augment-50K	noise study	40K/10K	balanced	zero shared links/samples; 97 shared maps

TABLE XIX: Field-specific numerical tolerance accuracy on the balanced MAIN-50K-v1 evaluation split (mean $\pm$ sample standard deviation over three seeds).

Field and tolerance	Accuracy ↑
Path count (± 1)	0.9533 ± 0.0061
First-path delay (± 50 ns)	0.9451 ± 0.0032
First-path angle ($\pm 15^\circ$)	0.6841 ± 0.0102
First-path power (± 5 dB)	0.6436 ± 0.0430
K factor (± 3 dB)	0.7971 ± 0.0302
Delay spread (± 50 ns)	0.9700 ± 0.0003
Angular spread ($\pm 15^\circ$)	0.7907 ± 0.0716
LoS delay (± 50 ns)	0.9915 ± 0.0012
LoS angle ($\pm 15^\circ$)	0.9972 ± 0.0007
Reflection-interaction count (± 1)	0.6562 ± 0.0031

TABLE XX: Complete aggregate performance of CSI2TEXT-PGMT-MAIN-50K on the balanced main split (mean $\pm$ sample standard deviation over three seeds).

Target	Result
Environment accuracy (%) ↑	100.00 ± 0.00
LoS/NLoS accuracy (%) ↑	99.58 ± 0.11
Path-count MAE ↓	0.215 ± 0.056
First-path delay MAE (ns) ↓	21.91 ± 0.30
First-path angle MAE ($^\circ$) ↓	16.61 ± 0.56
First-path power MAE (dB) ↓	4.34 ± 0.27
K-factor MAE (dB) ↓	2.15 ± 0.04
Delay-spread MAE (ns) ↓	11.30 ± 0.26
Angular-spread MAE ($^\circ$) ↓	9.72 ± 0.47
LoS-delay MAE (ns) ↓	8.83 ± 0.55
LoS-angle MAE ($^\circ$) ↓	1.33 ± 0.17
NLoS first-delay MAE (ns) ↓	40.66 ± 1.58
NLoS first-angle MAE ($^\circ$) ↓	35.88 ± 0.96
Reflection-interaction-count MAE ↓	1.032 ± 0.019
Reflected-path-count MAE ↓	0.086 ± 0.005

APPENDIX C
ADDITIONAL EXPERIMENTAL EVIDENCE

This section adds protocols and evidence not already tabulated in the main paper. Main-text AWGN, cross-configuration, split-sensitivity, propagation-regime, and language-alignment tables are not repeated.

A. Complete Main-Model Aggregate Performance

Table XX restores the full-label view from the initial draft. The first-path delay is aligned to the current MAIN-50K-v1 result, 21.91 ns; it should not be confused with the separately sampled COST-50K-RPATH-v2 value reported in the inference-cost subsection.

The categorical fields are nearly saturated and path-count error remains small, but the complete view reveals a pronounced propagation-regime asymmetry: LoS delay and angle errors are 8.83 ns and 1.33°, whereas the corresponding NLoS values are 40.66 ns and 35.88°. The two reflection quantities are intentionally separate: one counts encoded reflection events, while the other counts paths containing at least one reflection. The NLoS delay/angle entries were derived seed-wise on this exactly balanced cohort from $E_{\mathrm{NLoS}} = 2E_{\mathrm{all}} - E_{\mathrm{LoS}}$; this identity must not be reused for an imbalanced cohort.

B. Cross-Configuration Protocol and Extended Fields

The frequency-transfer model is trained jointly on $N_f \in \{64, 96, 192, 256\}$ and evaluated on $N_f = 128$, which is excluded from fitting. This tests interpolation to a held-out sampling configuration after phase-preserving resampling; it does not establish arbitrary transfer across carrier frequency or bandwidth. The mixed-array model is trained on UPA-8×8 and ULA-64 and evaluated on aligned held-out UPA-4×4, UPA-16×4, and ULA-32 groups. Directional ULA transfer is evaluated separately in both $64 \to 32$ and $32 \to 64$ directions. The complete per-setting results are summarized in the main paper; the principal protocol safeguard is that held-out array groups do not reuse fitted physical links.

C. Paired Multi-Numerology Expert Fusion

The main cross-configuration experiment asks whether one CSI observation acquired under an unseen frequency-sampling configuration can be interpreted without configuration-specific fine-tuning. We additionally consider an assisted setting in which two CSI observations of the same physical link are available under different numerologies. This setting represents multi-numerology sounding or coordinated measurements in which complementary frequency resolutions provide different evidence about the same propagation channel. Because both

TABLE XXI: Paired multi-numerology delay fusion from the same expert-gating experiment. Only fusion outputs are shown. Lower is better, and underlining marks the best fusion result in each column.

Fusion method	96+128 first delay ↓	96+128 LoS delay ↓	128+192 first delay ↓	128+192 LoS delay ↓
Direct fusion	63.14	32.05	49.79	26.17
A/B hard gate	63.31	24.70	32.73	19.90
Hard expert gate	60.45	25.04	35.28	21.69
Soft expert gate	57.58	23.24	37.67	20.53

observations are available at inference time, this experiment complements rather than replaces the single-input held-out-N_f evaluation.

Let $(\mathbf{H}^{(a)}, \mathbf{c}^{(a)})$ and $(\mathbf{H}^{(b)}, \mathbf{c}^{(b)})$ denote the paired observations. The shared encoder produces configuration-aware representations $\mathbf{z}_c^{(a)}$ and $\mathbf{z}_c^{(b)}$, from which the two experts generate structured predictions $\hat{\mathbf{y}}^{(a)}$ and $\hat{\mathbf{y}}^{(b)}$. A lightweight gate assigns sample-dependent continuous weights,

$$\big[g_a, g_b\big] = \mathrm{softmax}\Big(G\Big[\mathbf{z}_c^{(a)}; \mathbf{z}_c^{(b)}; \mathbf{c}^{(a)}; \mathbf{c}^{(b)}\Big]\Big), \qquad g_a + g_b = 1,$$

and forms the fused record

$$\hat{\mathbf{y}}_{\mathrm{soft}} = g_a \hat{\mathbf{y}}^{(a)} + g_b \hat{\mathbf{y}}^{(b)}.$$

Unlike unconditional direct fusion, the gate can vary the contribution of each numerology for every link. We compare soft weighting with hard expert selection and an A/B hard gate. The optional gate changes only the structured physical prediction; relational correction and deterministic verbalization remain unchanged.

For the $N_f = 96 + 128$ pair, soft expert gating reduces first-delay MAE from 63.14 to 57.58 ns and LoS-delay MAE from 32.05 to 23.24 ns relative to direct fusion. Averaged over the four pair–target entries in Table XXI, its MAE is 34.76 ns, compared with 42.79 ns for direct fusion, 35.16 ns for the A/B hard gate, and 35.62 ns for hard expert selection. Soft weighting therefore provides the strongest overall balance among the evaluated fusion strategies.

The improvement is not uniform. For $N_f = 128 + 192$, the A/B hard gate gives the lowest fused first-delay and LoS-delay errors, while soft weighting remains close. More generally, paired fusion cannot guarantee improvement when one numerology already provides substantially stronger evidence than the other. The result supports adaptive multi-numerology fusion as an optional extension under paired-input availability, rather than as a requirement of the main CSI-to-text pipeline. A separately trained calibrated confidence-fallback variant reduces signed prediction bias, but it is retained as a calibration diagnostic and is not mixed into this controlled expert-gating comparison.

D. Language Alignment and Relational Correction

The matched language-alignment results retained in the main paper use the current reported values; the obsolete Appendix J table from the earlier draft is intentionally not reproduced. All ablation variants retain direct physical supervision and the same deterministic verbalizer, so removing semantic alignment need not cause a general collapse. The supported conclusion is localized: alignment improves selected semantic and reflection-structure attributes but is not uniformly beneficial for every fully supervised regressand.

TABLE XXII: Language-alignment ablation under scenario-disjoint evaluation. Lower is better for MAE; higher is better for reflection accuracy. Underlining marks the best value in each row.

Metric	Full	No CSI–text	No text–prototype	Physics only
First-delay MAE (ns) ↓	39.3698	39.8403	40.9318	40.1218
First-delay LoS MAE (ns) ↓	19.8347	20.9726	22.1455	21.4466
First-delay NLoS MAE (ns) ↓	58.0109	58.8105	59.8201	58.8984
LoS-delay MAE (ns) ↓	17.6445	20.2644	19.3338	21.4869
LoS-angle MAE (°) ↓	2.2555	2.3383	2.3435	3.4443
First-angle LoS MAE (°) ↓	3.7392	3.9504	3.9118	4.3980
First-angle NLoS MAE (°) ↓	41.4775	40.6324	42.2838	40.3674
Path-count MAE ↓	0.1935	0.2051	0.2505	0.3267
Reflection accuracy ↑	0.5388	0.5213	0.5310	0.5314
Delay-spread MAE (ns) ↓	17.7460	17.9922	18.2041	17.8470
K-factor MAE (dB) ↓	3.3551	3.3279	3.3932	3.2915
First-power MAE (dB) ↓	5.5425	5.4669	5.5787	5.3613
Azimuth-spread MAE (°) ↓	8.7741	9.6889	8.1116	7.8690

The scenario-disjoint results localize the generalization benefit. Full alignment is best for all reported delay quantities, LoS geometry, path count, reflection accuracy, and delay spread. Relative to physics-only training, it reduces LoS-delay MAE by 17.9%, LoS-angle MAE by 34.5%, and path-count MAE by 40.8%. Text–prototype removal produces the largest delay degradation among the two pairwise removals, indicating that coarse propagation prototypes help preserve temporal structure outside the training scenarios. The result is not uniform: physics-only training remains better for NLoS first-angle, K factor, first-path power, and azimuth spread. Language alignment therefore acts as a structured semantic regularizer, not as a universal replacement for target-specific physical supervision.

Across every tested SNR, Full gives the lowest first-delay and first-angle MAE. At 30, 20, and 10 dB, its first-delay error is respectively 15.2%, 16.4%, and 16.9% lower than physics-only training. Removing text–prototype alignment causes a larger delay penalty than removing CSI–text alignment over most noisy conditions, supporting a robustness role for coarse propagation prototypes. Full alignment also gives the lowest reflection MAE from clean through 10 dB, although physics-only training becomes better at 0 dB. Power, K, and aggregate numerical tolerance accuracy retain attribute-specific trade-offs, and physics-only training has the highest reported numerical accuracy. The robust claim is therefore localized: semantic alignment stabilizes temporal, angular, and reflection-structure representations under corruption, but does not improve every raw-scale or tolerance-aggregated quantity.

The LoS identity $\tau_1 = \tau_{\mathrm{LoS}}$ follows from the ray-tracing semantics. An audit of 84,464 LoS links from ten source files found exact equality at 10^{-6} ns tolerance. On a representative seed-0 correction cohort, the raw first-path and LoS-delay heads have MAEs of 7.9958 and 6.4781 ns. Replacing both emitted slots with the dedicated LoS-delay estimate gives 6.4781 ns for both fields on the same mask, reducing emitted first-delay error by 1.5177 ns while leaving LoS-delay error unchanged. This is a post-processing diagnostic, not evidence that the raw neural head improved: known identities can remove contradictions,

TABLE XXIII: Language-alignment ablation under clean-to-moderate complex AWGN. Lower is better for physical MAE; higher is better for numerical tolerance accuracy. Uncertainty is shown where supplied.

Condition	Model	First delay ↓	First angle ↓	First power ↓
Clean	**Full**	25.974 ± 0.344	18.844 ± 0.404	5.116 ± 0.349
	No CSI–text	27.319	20.366	5.155
	No text–prototype	27.916	20.706	4.876
	Physics only	27.181	19.980	4.788
30 dB	**Full**	72.697 ± 0.401	23.101 ± 0.844	4.564 ± 0.004
	No CSI–text	77.509 ± 0.235	24.548 ± 0.052	4.888 ± 0.003
	No text–prototype	83.644 ± 0.090	24.619 ± 0.009	4.862 ± 0.014
	Physics only	85.775 ± 0.240	24.409 ± 0.088	4.567 ± 0.002
20 dB	**Full**	86.112 ± 0.960	24.292 ± 0.711	4.351 ± 0.002
	No CSI–text	89.023 ± 0.289	25.445 ± 0.046	4.745 ± 0.003
	No text–prototype	95.650 ± 0.413	26.076 ± 0.065	4.787 ± 0.372
	Physics only	103.020 ± 0.319	25.495 ± 0.065	4.461 ± 0.006

Condition	Model	K factor ↓	Refl. MAE ↓	Numeric acc. ↑
Clean	**Full**	2.084 ± 0.124	1.018 ± 0.014	0.7405 ± 0.0057
	No CSI–text	2.545	1.091	0.8360
	No text–prototype	1.928	1.083	0.8438
	Physics only	2.036	1.091	0.8477
30 dB	**Full**	2.142 ± 0.125	1.180 ± 0.006	0.6927 ± 0.0065
	No CSI–text	2.342 ± 0.008	1.201 ± 0.000	0.7972 ± 0.0006
	No text–prototype	1.930 ± 0.006	1.192 ± 0.003	0.7976 ± 0.0003
	Physics only	2.101 ± 0.004	1.202 ± 0.002	0.8053 ± 0.0010
20 dB	**Full**	2.612 ± 0.118	1.232 ± 0.011	0.6491 ± 0.0082
	No CSI–text	2.676 ± 0.010	1.240 ± 0.004	0.7579 ± 0.0006
	No text–prototype	3.070 ± 0.009	1.307 ± 0.004	0.7387 ± 0.0006
	Physics only	2.335 ± 0.006	1.250 ± 0.002	0.7746 ± 0.0007

TABLE XXIV: Language-alignment ablation under severe complex AWGN. Metric conventions match Table XXIII.

Condition	Model	First delay ↓	First angle ↓	First power ↓
10 dB	**Full**	100.231 ± 1.727	25.123 ± 0.786	5.377 ± 0.576
	No CSI–text	101.013 ± 0.212	25.778 ± 0.055	4.814 ± 0.015
	No text–prototype	107.434 ± 0.269	30.381 ± 0.046	5.248 ± 0.023
	Physics only	120.664 ± 0.467	27.374 ± 0.089	4.760 ± 0.007
0 dB	**Full**	209.288 ± 12.315	32.296 ± 1.919	17.692 ± 0.076
	No CSI–text	209.599 ± 0.669	33.194 ± 0.115	18.059 ± 0.053
	No text–prototype	222.284 ± 0.419	35.155 ± 0.145	17.598 ± 0.005
	Physics only	209.656 ± 0.610	37.217 ± 0.077	15.375 ± 0.021

Condition	Model	K factor ↓	Refl. MAE ↓	Numeric acc. ↑
10 dB	**Full**	5.494 ± 0.582	1.326 ± 0.009	0.5622 ± 0.0101
	No CSI–text	5.874 ± 0.018	1.332 ± 0.006	0.6845 ± 0.0008
	No text–prototype	10.128 ± 0.034	1.344 ± 0.003	0.6425 ± 0.0004
	Physics only	6.322 ± 0.005	1.427 ± 0.002	0.7036 ± 0.0001
0 dB	**Full**	19.855 ± 0.659	2.384 ± 0.178	0.3393 ± 0.0218
	No CSI–text	21.525 ± 0.052	2.446 ± 0.002	0.4684 ± 0.0004
	No text–prototype	21.946 ± 0.004	2.412 ± 0.003	0.4702 ± 0.0005
	Physics only	20.390 ± 0.009	2.211 ± 0.003	0.5005 ± 0.0003

but cannot repair unrelated physical quantities.

TABLE XXV: Physical quality on the inference-cost cohort. Decoder first-delay values can differ from the learned-baseline table because this benchmark uses a different cohort; model suffixes make this explicit.

Method	Delay (ns) ↓	Angle (°) ↓	Power (dB) ↓	K (dB) ↓
CSI2TEXT-PGMT-COST-BENCHMARK	20.68	19.55 ± 0.38	5.02 ± 0.31	1.87 ± 0.24
Single-task	47.75 ± 5.22	20.65 ± 0.32	3.85 ± 0.23	1.42 ± 0.04
PDP/IFFT+MLP	27.58 ± 5.05	41.62 ± 0.68	12.66 ± 0.32	5.22 ± 0.22
Flattened MLP	109.98 ± 0.87	34.85 ± 0.26	13.06 ± 1.17	5.28 ± 1.02
Transformer	265.91 ± 0.02	53.37 ± 0.02	38.18 ± 0.00	22.52 ± 0.00
CNN	1619.16 ± 1238.13	102.18 ± 42.31	14595.56 ± 18584.48	55.55 ± 34.37
Qwen3-1.7B	177.66	42.84	6.17	2.56
Qwen3.5-2B	162.47	42.37	5.85	2.53
DeepSeek-8B	232.66	59.16	27.77	16.42
InternLM3-8B	165.56	40.22	5.96	2.49

E. Selective Prediction and Uncertainty Coverage

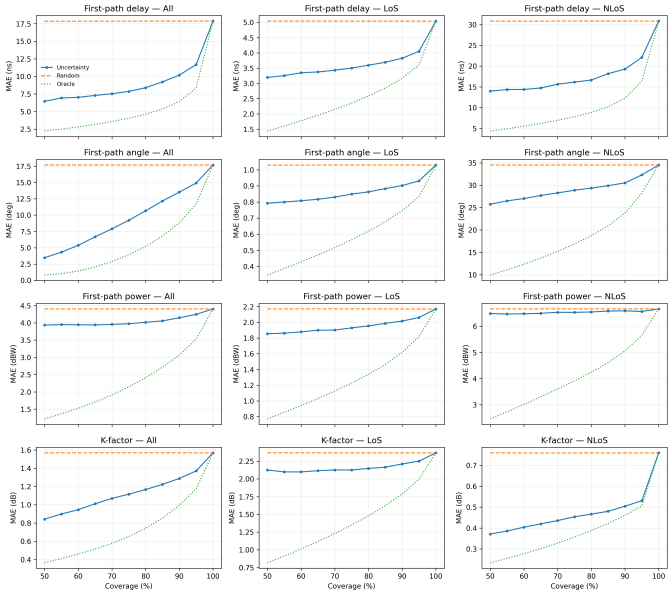


Fig. 9: Risk–coverage curves from the three-seed ensemble for first-path delay, angle, power, and K-factor, evaluated jointly and by LoS/NLoS regime. Blue curves rank samples by ensemble disagreement, orange lines show random rejection, and green curves give the unattainable oracle ranking by realized error. Lower coverage means that more high-uncertainty reports are deferred.

Figure 9 expands the selective-prediction analysis in Figure 7. Ensemble disagreement is informative for delay and angle: their MAEs fall steadily as high-uncertainty samples are removed, with the largest absolute benefit in the NLoS regime. Power is less separable, especially for NLoS, indicating that seed disagreement does not capture every raw-scale failure. K-factor improves more consistently, although its LoS and NLoS scales differ. The remaining gap to the oracle curves quantifies room for better uncertainty estimation. These curves support selective deployment rather than replacement of the full-coverage main metrics: deferred samples require another CSI observation, multi-numerology fusion, or review.

F. Inference Cost

The runtime and memory measurements are consolidated in Table III. Latency is measured end to end on 200 common examples; quality uses the 1,000-example decoder cohort.

Conventional models use FP32 and direct decoders use 4-bit NF4 weights with BF16 computation. The structured pipeline needs 13.58 ms and 0.13 GB, versus 11–23 s and 1.53–6.18 GB for autoregressive decoders. This is a deployment advantage rather than a replacement for the physical and text measurements.

For completeness, the original Pareto visualization normalizes each continuous MAE as

$$q_m = \frac{1}{1 + \mathrm{MAE}_m / T_m}, \tag{16}$$

where T_m is the field-specific tolerance. The five shared physical fields are

$$\mathcal{M}_A = \{\tau_1, \theta_1, p_1, K, n_{\mathrm{refl}}\}.$$

For this set,

$$S_{\mathrm{phys}} = 100 \left(\prod_{m \in \mathcal{M}_A} q_m \right)^{1/5}. \tag{17}$$

For the eight-field continuous set $\mathcal{M}_C$—first and LoS delay, delay spread, first and LoS angle, angular spread, first-path power, and K factor—the score is

$$S_{\mathrm{cont}} = 100 \left(\prod_{m \in \mathcal{M}_C} q_m \right)^{1/8}. \tag{18}$$

The structural and text components are

$$S_{\mathrm{struct}} = 100 \left(a_{\mathrm{LoS}} a_{\mathrm{path}} a_{\mathrm{refl}} a_{\mathrm{rpath}} \right)^{1/4}, \tag{19}$$

$$S_{\mathrm{text}} = 100 \left(F A_{\mathrm{num}} F_{\mathrm{slot}} (1 - H) A_{\mathrm{delay}} \right)^{1/5}, \tag{20}$$

and the complete score is

$$S_{\mathrm{complete}} = 0.50 S_{\mathrm{cont}} + 0.20 S_{\mathrm{struct}} + 0.30 S_{\mathrm{text}}. \tag{21}$$

Here a_{LoS} averages visibility-state accuracy and macro-F1; a_{path} and a_{rpath} are within-one count accuracies; a_{refl} is reflection-bin accuracy; F is factuality; A_{num} is numerical tolerance accuracy; F_{slot} is slot F1; H is hallucination rate; and A_{delay} is delay tolerance accuracy. Rule consistency is deliberately excluded. For CSI2TEXT-PGMT-COST-BENCHMARK, the original visualization gives $S_{\mathrm{complete}} = 78.28 \pm 0.23$, including $S_{\mathrm{struct}} = 87.34 \pm 0.18$ and $S_{\mathrm{text}} = 94.53 \pm 0.19$. Geometric means penalize a single failed field, but the tolerances and the 0.50/0.20/0.30 weights are design choices that can change rankings. The composite is therefore a visualization aid, not a replacement for the raw measurements.

G. Reflected-Path Extension

The main record predicts total reflection-interaction count. A separate head extends the record with the number of distinct reflected paths. The extension uses RPATH-100K, a naturally imbalanced 100,000-example cohort containing 13,628 LoS and 86,372 NLoS links. Its 80,000/20,000 split contains 10,902/69,098 LoS/NLoS training examples and 2,726/17,274 test examples, so it is not pooled with the balanced main comparison.

The verbalized reflected-path count reaches MAE 0.091 ± 0.007, RMSE 0.234 ± 0.008, and within-one accuracy $98.82 \pm 0.06\%$. Complete descriptions reach $94.06 \pm 1.07\%$ factuality, $99.974 \pm 0.012\%$ numeric-slot F1, $0.0437 \pm 0.0200\%$ hallucination, $70.63 \pm 5.48\%$ numerical tolerance accuracy, and $99.872 \pm 0.029\%$ rule consistency. The result demonstrates that total reflection multiplicity and the number of reflection-bearing paths can be modeled separately. Its strong NLoS skew precludes a direct numerical comparison with the balanced main cohort.

Appendix D
Qualitative Results, Failure Cases, and Limitations

A. Representative Channel-Report Templates

The deterministic verbalizer maps each corrected field to one auditable sentence slot. A LoS report contains the predicted environment and visibility state, retained path count, first-path delay, angle and power, delay and angular spreads, K factor, LoS-specific delay and angle, total reflection-interaction count, and reflected-path count when the extension is enabled. An NLoS report suppresses LoS-only slots. The corresponding templates are

$$\begin{aligned} \text{LoS: } & [e, \mathrm{LoS}, n_p, \tau_1, \theta_1, p_1, \sigma_\tau, \sigma_\theta, \\ & K, \tau_{\mathrm{LoS}}, \theta_{\mathrm{LoS}}, n_{\mathrm{refl}}, n_{\mathrm{rpath}}], \\ \text{NLoS: } & [e, \mathrm{NLoS}, n_p, \tau_1, \theta_1, p_1, \sigma_\tau, \sigma_\theta, \\ & K, n_{\mathrm{refl}}, n_{\mathrm{rpath}}]. \end{aligned}$$

Qualitative inspection compares every emitted number with the corresponding corrected field rather than judging surface fluency alone. A reproducible case audit records the sample and cohort ID, reference record, pre-correction prediction, corrected record, final description, and field-level error.

B. Structured Model Versus Direct Decoder Audits

The direct decoders usually master the output schema: their slot F1 values are close to those of the structured model. Their main failure is numerical grounding, especially delay. A useful qualitative comparison must therefore mark separately: missing slots, unsupported slots, internally inconsistent values, and values that are syntactically valid but outside physical tolerance. This distinction prevents perfect rule consistency from being misread as perfect channel prediction.

C. Before and After Relational Correction

Relational correction operates only on the emitted record. For predicted LoS samples, the earliest-path and direct-LoS delays are reconciled to the dedicated LoS-delay estimate; negative spreads and invalid counts are clamped; and at least one retained path is enforced. The raw head outputs remain the proper measure of neural prediction quality. Before/after examples should therefore display both records and identify which fields were changed by deterministic rules.

D. LoS and NLoS Failure Cases

The dominant failure regime is NLoS first-path inference. Direct-path geometry produces stable delay and angular signatures, while an indirect first retained path is more weakly constrained by the supplied transmitter-side observation. Failure analysis should be stratified into: incorrect first-path delay, incorrect receiver-side angle, absolute-power error, reflection-count error, and schema-level error. The main results show that schema errors are rare relative to physical estimation errors. Future qualitative panels should sample both typical and high-error cases from fixed cohort IDs rather than selecting only visually favorable outputs.

E. Scope and Limitations

All labels and physical relations are derived from finite ray-tracing exports. They inherit the simulator's path-retention policy, fixed transmit-power assumption, maximum retained-path count, and reflection encoding. The work establishes a controlled simulator-grounded interface, not field deployment. Measured-channel validation remains necessary.

The language layer demonstrates auditable realization of a typed record rather than unrestricted conversation or stylistic generation. Composite factuality is slot based and tolerance dependent; raw per-field errors must remain visible. Generalization covers selected frequency and array shifts, not arbitrary bandwidths, carriers, calibration errors, non-Gaussian interference, or hardware impairments. The AWGN experiment probes one controlled corruption axis only.

The direct decoders have one run each and are not parameter matched, so small ranking differences are unresolved. Multi-task sharing and language alignment do not improve every target. The robust conclusion is narrower: target-appropriate representations and structured prediction ground reports better than direct decoding, while NLoS multipath recovery remains the principal limitation.